



Efficient and adaptive sensory codes

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The ability to adapt to changes in stimulus statistics is a hallmark of sensory systems. Here, we developed a theoretical framework that can account for the dynamics of adaptation from an information processing perspective. We use this framework to optimize and analyze adaptive sensory codes, and we show that codes optimized for stationary environments can suffer from prolonged periods of poor performance when the environment changes. To mitigate the adversarial effects of these environmental changes, sensory systems must navigate tradeoffs between the ability to accurately encode incoming stimuli and the ability to rapidly detect and adapt to changes in the distribution of these stimuli. We derive families of codes that balance these objectives, and we demonstrate their close match to experimentally observed neural dynamics during mean and variance adaptation. Our results provide a unifying perspective on adaptation across a range of sensory systems, environments, and sensory tasks.

The natural sensory environment is constantly in flux. When environmental conditions change, so too do the statistics of sensory stimuli. In response to these changes, sensory systems modify how incoming stimuli are encoded in neural response patterns. This phenomenon, known as sensory adaptation, has been studied since the beginnings of sensory neuroscience¹.

Adaptation occurs across sensory systems, regardless of the sensory modality or stage along the ascending sensory pathway. Adaptation has been observed in the visual², auditory^{3,4}, somatosensory⁵, olfactory⁶, and electrosensory⁷ systems. It is manifested in individual neurons^{8,9} and neural populations^{4,10,11}, beginning in the sensory periphery^{2,12} through the midbrain^{13,14} to the cortex^{11,15}. Although the most extensively studied forms of adaptation are induced by changes in low-order stimulus statistics, such as mean or variance^{4,8}, adaption has been observed to more complex statistics, such as the spatial scale¹⁶ and high-order structure¹⁰ of visual stimuli. Despite the prevalence of adaptation across the brain, it remains unclear as to whether this diverse range of phenomena emerges from a common set of neural coding principles.

A dominant hypothesis is that adaptation reflects a dynamic reallocation of finite metabolic resources to support the performance of different sensory tasks^{8,9,17}. As postulated by the efficient coding hypothesis¹⁸, sensory neurons might be adapted to stimulus statistics in a manner that maximizes the information transmitted downstream. Different variants of this hypothesis propose that sensory systems might only need to encode specific features of incoming sensory signals, depending on the task^{19,20}. Regardless of the task, however, this hypothesis dictates that the neural code must be adapted to the stimulus distribution to efficiently encode and transmit task-relevant information. As such, this hypothesis accounts for the consequences of sensory adaptation: to maintain a given level of task performance, sensory neurons should modify their encoding properties when the distribution of incoming stimuli changes, and efficient coding specifies what these properties should be when the environment is stationary and the sensory code is adapted. It does not, however, specify how the modification of these properties should be accomplished nor whether this modification necessitates different coding strategies than would be required in stationary settings.

In this study, we extended the efficient coding framework to non-stationary environments and to adaptive sensory representations,

and we show that, in certain scenarios, adaptive codes must use different strategies than non-adaptive ones. We consider sensory systems that use an internal estimate of the current sensory context, specified by the distribution of incoming stimuli, to guide adaptation^{21,22}. When the environment is stationary, this estimate can, in principle, be used to design a code that maximizes task performance, as posited by classic theories of efficient coding^{12,23,24}. However, when the environment changes, mismatches between this estimate and the current context can induce substantial errors. Optimal adaptive codes must therefore devote some of their limited resources to either detecting changes in the underlying distribution of stimuli or reducing coding errors induced by an incorrect estimation of the underlying context. We extend the efficient coding framework to formalize this interplay and derive codes that optimally balance these competing objectives. This extended framework generalizes current theories of neural computation to non-stationary environments and, in doing so, enables a comparison of adaptation schemes across sensory representations, environments, and tasks.

Results

Consider a sensory neuron that encodes local image features across different visual scenes (Fig. 1a). The stochastic response of this neuron can be captured by a linear filter applied to individual image patches, followed by a saturating non-linearity that transforms noisy filter outputs into Poisson-distributed firing rates²⁵. To accurately decode filter outputs, the saturating non-linearity must be adapted to the distribution of these outputs^{9,12,20,26,27}.

During visual exploration of a natural scene, the distribution of these filter outputs will vary (Fig. 1b)¹⁴. This variability can be summarized by a set of distributions that represent different sensory contexts and that span a range of local luminance (mean) and contrast (variance) values observed across an ensemble of natural scenes (Fig. 1c, lower panels).

To minimize decoding error, the non-linearity might be globally adapted to the marginal distribution of filter outputs averaged across all contexts (Fig. 1c, upper left). A more advantageous strategy requires the neuron to locally adapt its non-linearity to match each individual context (Fig. 1c, upper right), such that its limited dynamic range is devoted to stimuli encountered in that context (Fig. 1c, lower panels)^{8,9,14}. This local adaptive strategy reduces

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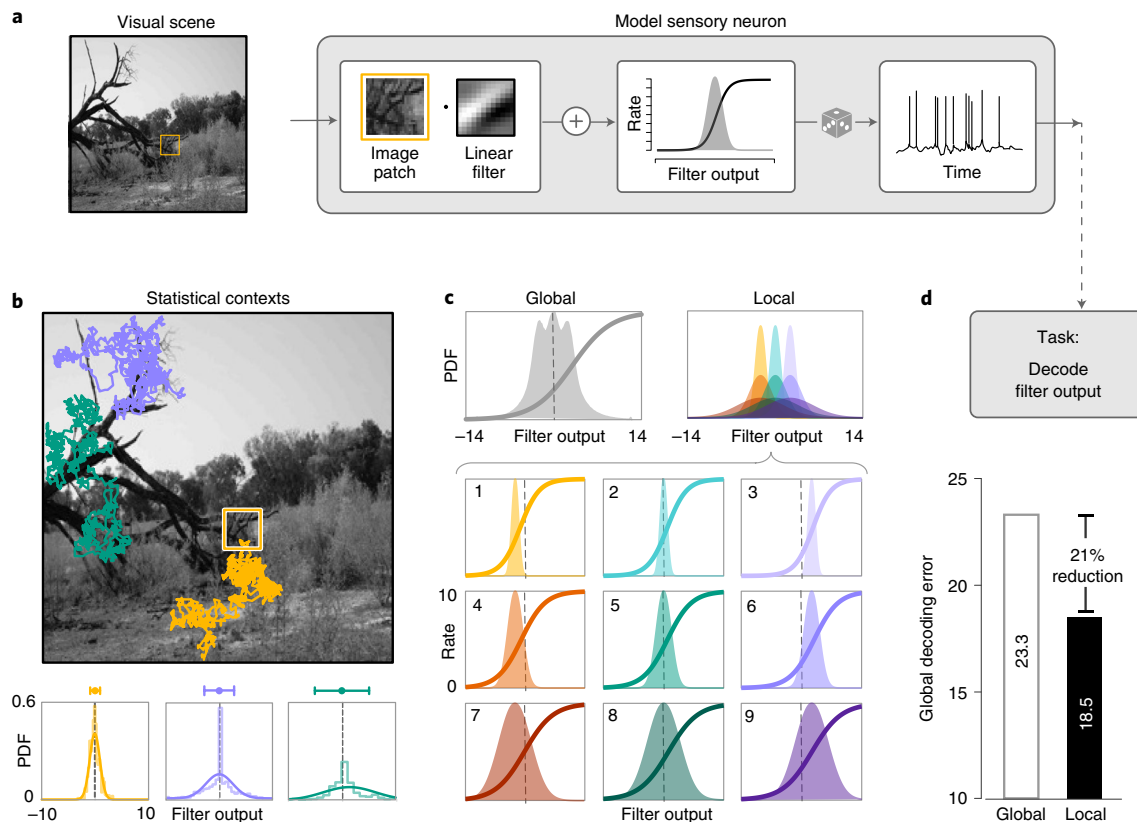


Fig. 1 | Adaptation can improve performance on sensory tasks. a, A linear-non-linear Poisson neuron encodes image patches from different regions of a visual scene (image from the UPenn Image Database; Methods). Local image patches are first filtered with a linear receptive field. The output of this filter is then distorted by noise, passed through a saturating non-linearity, and used to generate a spiking response via a Poisson process. The saturating non-linearity can be optimized for the task of decoding the filter output from the spiking response of the neuron. Global decoding error is computed as the mean squared error between the ML estimate and the true value of the filter output. **b**, Visual exploration of different regions of a visual scene (colored lines, main panel) results in different distributions of the filter output (lower panels, shown as probability density functions (PDFs)). Histograms, empirical distribution of filter outputs; error bars, empirical mean $\pm$ s.d.; solid lines, best-fitting Gaussian distribution. **c**, We approximate the variability in filter outputs by a set of nine Gaussian distributions that span a range of means and variances observed across an ensemble of natural scenes. A global non-linearity (gray line, upper-left panel) minimizes the average decoding error given the distribution of filter outputs marginalized over all nine contexts (shaded distribution). Local non-linearities (solid lines, lower panels) minimize the average decoding error given the distribution of filter outputs within each individual context (shaded distributions). **d**, When averaged over filter outputs drawn evenly from each of the nine contexts, the global decoding error is higher if the neuron uses a fixed global non-linearity for all contexts (white bar) than if it locally adapts the non-linearity to match each individual context (black bar).

decoding error well beyond what is achieved when the same neuron uses a global non-linearity adapted to all contexts (Fig. 1d).

A normative framework for adaptive sensory codes. This simple model neuron highlights the well-known advantages of adapting to context-dependent changes in input statistics^{9,14}. However, this form of adaptation assumes that the neuron uses perfect context knowledge to optimally adapt its non-linearity. In reality, sensory systems must infer context from incoming sensory stimuli and can, at best, use a context estimate to adapt. Because of errors inherent to any estimation process, there will be times when the system uses a suboptimal strategy that is not matched to the current input statistics (Fig. 2a). In what follows, we demonstrate that these periods of mismatch can cause considerable errors in neural coding. We then show how sensory systems can reduce these mismatch-induced errors and how this, in turn, can affect performance during times when the system is correctly matched to the input statistics.

To understand the interplay between these coupled sources of error, we consider an adaptive system that must perform a sensory

task in a changing environment (Fig. 2b). At a given time t , the system receives incoming stimuli s_t . The distribution of these stimuli, $s_t \sim p(s_t|c_t)$, is controlled by a latent context c_t that can change over time, with dynamics governed by $p(c_t|c_{t-1})$. The system transforms incoming stimuli into neural responses $r_t \sim p(r_t|s_t; \psi_t)$ using an adaptive encoder that has limited coding precision and that is controlled by encoding parameters ψ_t selected from a set Ψ . These neural responses are then used to perform a sensory task whose instantaneous error $e_t(r_t) \equiv e(s_t, r_t; \psi_t)$ depends on how incoming sensory stimuli are encoded in neural responses via the encoding parameters ψ_t . To improve task performance, the system can adapt the encoding parameters over time. To do this, an observer uses neural responses to construct a posterior belief $p(c_t|r_{t \leq t}; \psi_t)$ about the current context. This belief can then be used to construct a point estimate $\hat{c}_t$ of the current context and a point prediction of the future context. This prediction can then be used to adapt the upstream encoding parameters via a feedback loop. In practice, the current estimate provides a good approximation of this prediction in scenarios where the context changes with a small probability or by a small amount per unit time; we consider such scenarios for the remainder of the paper.

We evaluate the system's overall performance by averaging the instantaneous task error over ongoing changes in context. If the system spends a total time T_c in each of N_c different contexts, this global (that is, context-averaged) error can be written as:

$$E(\Psi) = \frac{1}{T} \sum_{t=1}^T e_t(\psi_t) = \frac{1}{T} \sum_{c=1}^{N_c} \sum_{t=1}^{T_c} e_t(\psi_t) = \frac{1}{T} \sum_{c=1}^{N_c} T_c E_c(\Psi) \quad (1)$$

where $E_c(\Psi) = (1/T_c) \sum_{t=1}^{T_c} e_t(\psi_t)$ is a local (that is, context-specific) error rate.

To achieve low global error, the encoding parameters $\psi_t \in \Psi$ should be adapted to the statistics of incoming stimuli. In the simplest case, exemplified in Fig. 1c, the system can use a fixed set of parameters $\Psi \equiv \Psi^{\text{global}}$ that are chosen to minimize the global error in Equation (1) and are thereby adapted to the set of all contexts:

$$\Psi^{\text{global}} = \arg \min_{\Psi} E(\Psi) \quad (2)$$

We refer to this as the 'static' code.

The system can further improve performance by adapting these parameters to individual contexts. In this case, the system can maintain a set of parameters $\Psi \equiv \{\psi_c^{\text{local}}\}$ that are optimal for each context c by minimizing the local error rate E_c . With perfect knowledge of the context c_t at a given time t , an 'oracle' code can then select among these context-optimal parameters, $\psi_t = \psi_{c=c_t}^{\text{local}}$. In the absence of perfect knowledge, the system can, instead, use a context estimate $\hat{c}_t$ to select among these same parameters, $\psi_t = \psi_{c=\hat{c}_t}^{\text{local}}$; we refer to this as the 'naively adaptive' code. Note that a code is specified both by the method of optimizing the parameters (here, minimizing either global or local error, as indicated by the superscript) and the method of selecting among these parameters (using no context knowledge, perfect context knowledge, or an estimate of context, as indicated by the subscript). These properties are summarized in Fig. 2c.

Because no real-world estimation strategy is perfect, any system that adapts based on a context estimate will encounter times when this estimate is correct and the system is matched to the current context ($\hat{c}_t = c_t$) and times when this estimate is incorrect and the system is mismatched ($\hat{c}_t \neq c_t$). This observation allows us to decompose the global error in Equation (1):

$$\begin{aligned} E(\Psi) &= \underbrace{\frac{1}{T} \sum_{t: c_t = \hat{c}_t} e_t(\psi_t)}_{\text{matched error}} + \underbrace{\frac{1}{T} \sum_{t: c_t \neq \hat{c}_t} e_t(\psi_t)}_{\text{mismatched error}} \\ &= \frac{1}{T} \left[\underbrace{(T - T_{\text{mis}}(\Psi))}_{\text{duration of match}} \times \underbrace{E_{\text{match}}(\Psi)}_{\text{matched error rate}} \right. \\ &\quad \left. + \underbrace{T_{\text{mis}}(\Psi)}_{\text{duration of mismatch}} \times \underbrace{E_{\text{mis}}(\Psi)}_{\text{mismatched error rate}} \right] \quad (3) \end{aligned}$$

where $(T - T_{\text{mis}})$ and T_{mis} denote the total duration of time that the system is 'matched' or 'mismatched', respectively, and E_{match} and E_{mis} denote the respective error rates during these times. Periods of mismatch can lead to high errors if either the duration of these mismatches is long or if error rates during periods of mismatch are high.

To date, theoretical studies of sensory coding have focused on the design of codes that are optimally matched to the current context and thus minimize the matched error rate E_{match} given

perfect context knowledge^{20,24,28–30}. Here, we extend these theories by observing that when the system does not have perfect knowledge of a time-varying context, its performance is determined by the interplay between the matched error rate and two additional factors: the mismatched error rate E_{mis} and the duration of mismatch T_{mis} . These three factors are intrinsically coupled and must therefore be balanced to maximize performance.

The naively adaptive code, which minimizes the matched error rate given the system's current context estimate, is the most direct extension of previous work. This code, which prioritizes local task performance given a current context estimate, suffers from high task errors when this estimate is faulty. To reduce these errors, we construct codes that minimize either the duration or the error rate of mismatches (Fig. 2c). The former prioritizes accurate inference of the underlying context (with parameters $\{\psi_c^{\text{infer}}\}$ selected via a context estimate, $\psi_t = \psi_{c=\hat{c}_t}^{\text{infer}}$); the latter prioritizes task performance in contexts other than the locally, but potentially incorrectly, estimated context (with parameters $\{\psi_c^{\text{nonlocal}}\}$ selected via a context estimate, $\psi_t = \psi_{c=\hat{c}_t}^{\text{nonlocal}}$). We interpolate between these and the naively adaptive code by constructing families of so-called 'biased' codes that balance these priorities. We index these codes with a bias parameter α_1 (for a code that prioritizes inference) or α_{NL} (for a code that prioritizes non-local task performance). These two parameters lie between 0 (for a fully local (that is, naively adaptive) code) and 1 (for a fully biased code).

To understand the interplay among different sources of task error, we evaluate how these families of codes balance different contributions to the global error in Equation (3); Fig. 2d,e exemplifies how we dissect these contributions (note that we report the relative duration of mismatch T_{mis}/T , which, when multiplied by E_{mis} , gives the total mismatched error). Although biased codes are designed to reduce either the duration or the error rate of mismatches, they are not guaranteed to reduce both simultaneously (Fig. 2d) and are therefore not guaranteed to reduce the overall mismatched error. Moreover, biased codes can only increase matched error over the naively adaptive code; the latter minimizes this error by construction (Fig. 2e, left). Although the precise balance of errors—and, consequently, the optimal code—can be shaped by factors such as number and diversity of statistical contexts, the measure of task error, and the nature of coding constraints (Supplementary Note), we highlight settings in which biased codes substantially reduce mismatched error, thereby overcoming increases in matched error and ultimately improving global performance over the naively adaptive code (Fig. 2e, right).

We measure this improvement relative to the difference in global error between the static and oracle codes, and we demonstrate that this improvement holds across a range of scenarios and even for simplified coding schemes. However, performance can be assessed via different measures, beyond global error. For example, rapid detection of environmental changes might require codes that minimize T_{mis} , whereas maintaining low instantaneous errors might require codes that minimize E_{mis} . By understanding how these different terms contribute to global error, we can dissect the properties of adaptive dynamics that reflect a range of coding objectives.

Locally suboptimal performance can improve global performance. We now revisit our linear–non-linear neuron (Fig. 1) through the lens of this framework (Fig. 3). Here, we explicitly model the temporal dynamics of the environment, which we take to switch between each of the nine different contexts at a small but fixed probability per time (Methods).

We approximate the distributions of filter outputs in each context as Gaussian distributions with context-dependent means and variances (that is, $p(s_t|c_t) = \mathcal{N}(s_t; \mu(c_t), \sigma(c_t)^2)$). At time t , a filter output s_t is sampled from the distribution specified by the context c_t and transformed into a Poisson-distributed spike count r_t via a

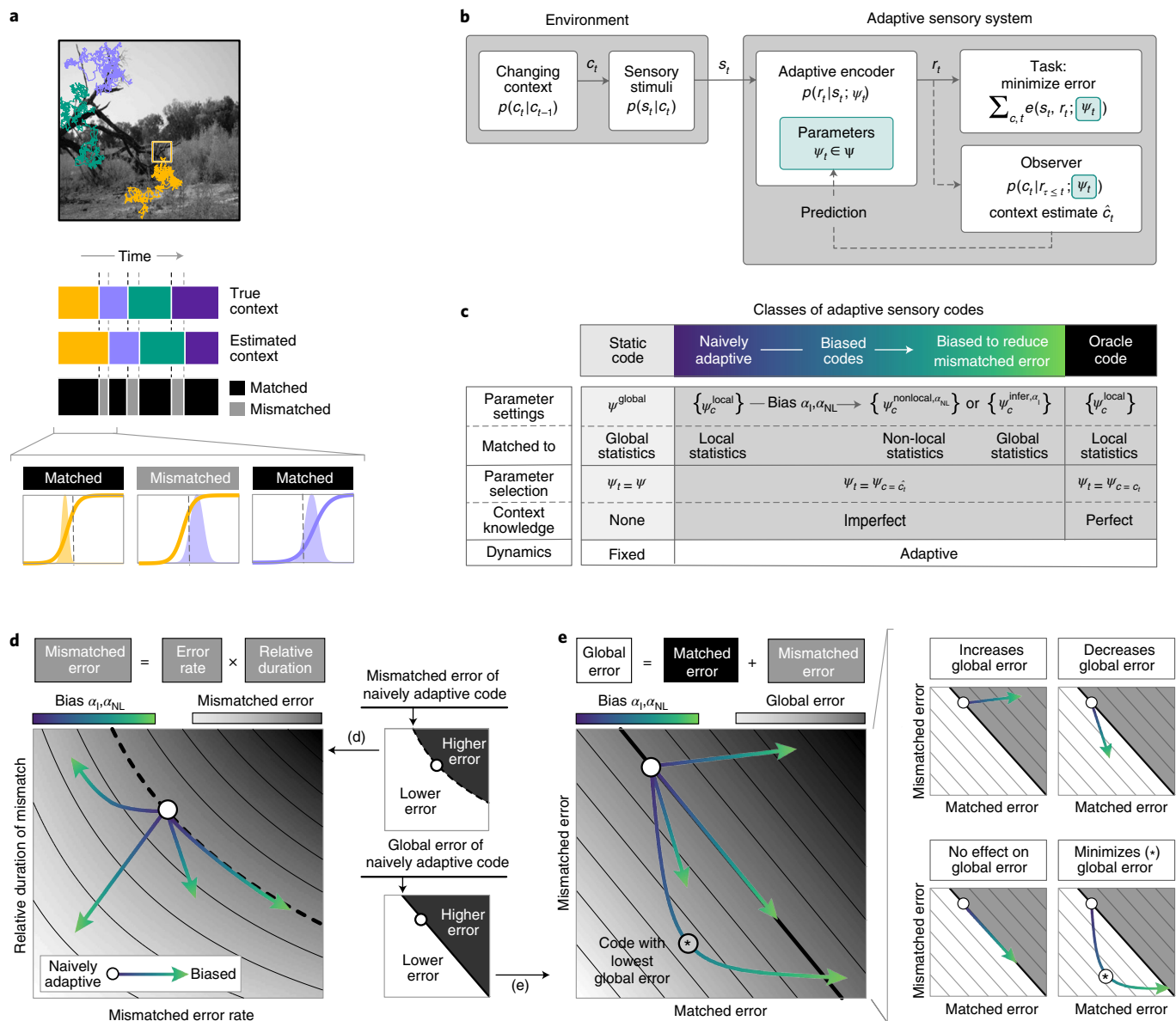


Fig. 2 | Adaptive sensory codes exhibit different patterns of error. **a**, A sensory system that must estimate a changing context (colored bars) will experience periods of time when this estimate is matched (black) or mismatched (gray) to the current context, illustrated for the example shown in Fig. 1. We design adaptive codes that can reduce the effect of these periods of mismatch. **b**, We consider a general scenario in which a changing context is signaled by incoming sensory stimuli that are transformed into neural responses by an adaptive encoder with parameters ψ (details in text). The output of this encoder is used to perform a sensory task. To improve task performance, the encoder can use an internal estimate of context to adapt its encoding parameters. **c**, We organize sensory codes according to whether their encoding parameters are fixed in time or adaptive and, if adaptive, whether they have perfect or imperfect knowledge of context. At one extreme, a 'static' code without context knowledge is matched to the distribution of stimuli across contexts; at the other extreme, an 'oracle' code with perfect context knowledge is matched to the distribution of stimuli within individual contexts. In between, we construct families of biased codes that have imperfect context knowledge and must balance errors that arise during matched and mismatched periods. **d**, Mismatched error (heat map) is the product between the mismatched error rate (horizontal axis) and the duration of mismatch (vertical axis). We begin with a naively adaptive code (open circle), and we bias this code toward decreasing either the error rate or the duration of mismatch. The resulting families of biased codes (colored arrows) can occupy different regions of this space. Solid lines, contours of constant mismatched error. Inset, upper right: the thick dashed line partitions the mismatched error surface into regions that achieve higher (gray) or lower (white) mismatched error than the naively adaptive code. **e**, The global error (heat map) is the sum of matched error (horizontal axis) and mismatched error (vertical axis). Solid lines, contours of constant global error. Inset, lower left: the thick solid line partitions the global error surface into regions that achieve higher (gray) or lower (white) global error than the naively adaptive code. Biased codes are guaranteed to have higher matched error than the naively adaptive code, but can have differing effect on global error (four small panels). We seek the code that achieves the lowest global error, marked by the star.

sigmoidal non-linearity with adaptable slope k and offset s_0 . The finite dynamic range of this non-linearity controls the maximum firing rate of the neuron and thus limits the precision with

which incoming stimuli can be encoded in the response r_t . Neural responses over a time window ΔT are then used to update a maximum likelihood (ML) estimate $\hat{c}_t$ of the current context, which is

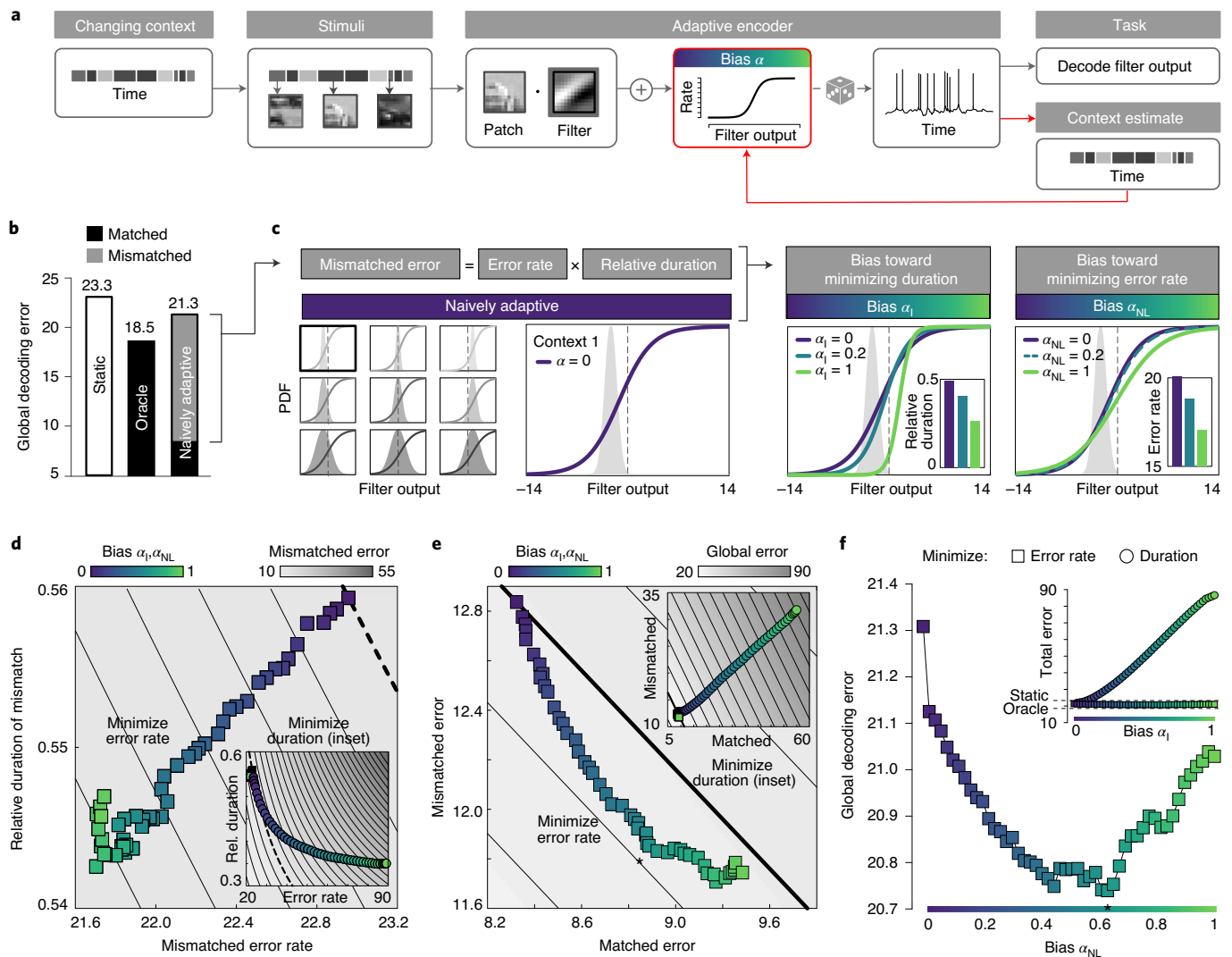


Fig. 3 | Biased codes improve global performance. **a**, We revisit the example Poisson neuron shown in Fig. 1 through the lens of the framework shown in Fig. 2. Here, the environment can switch between each of the nine contexts with a small but fixed probability per time. Rather than having perfect knowledge of this context, the neuron must construct an estimate of context that can be fed upstream (red arrow) and used to adapt the encoding non-linearity for the task of decoding filter outputs. **b**, A naively adaptive code that adapts based on a context estimate achieves performance intermediate between static and oracle codes. A large fraction of the global error arises during periods of mismatch (gray), when the context estimate is inaccurate. **c**, The naively adaptive code uses its context estimate to select among the context-optimal non-linearities shown in Fig. 1b and reproduced here. To reduce mismatched error, we bias these non-linearities toward inference (middle panel) or non-local task performance (right panel). The former is achieved by biasing each context-optimal non-linearity toward the non-linearity that maximally discriminates between all nine contexts (green curve), which reduces the duration of mismatch (inset). The latter is achieved by biasing each context-optimal non-linearity toward the non-linearity that accurately decodes stimuli in other contexts (green curve), which reduces the mismatched error rate (inset). **d**, In this scenario, codes biased toward inference (circles) result in relatively large mismatched error rates (inset). Codes biased toward non-local task performance (squares) show a reduction in both the duration and the error rate of mismatches and thereby achieve lower mismatched error (main panel). Solid lines, contours of constant mismatched error. **e**, Codes biased toward inference (circles) increase both matched and mismatched error (inset). Codes biased toward non-local task performance (squares) exhibit a tradeoff between matched and mismatched error (main panel). The code that optimally balances matched and mismatched error (and thereby achieves the lowest global error) is marked by a star. Solid lines, contours of constant global error. **f**, As bias increases, codes biased toward inference (circles) quickly exceed the global error produced by the static code and continue to monotonically increase thereafter (inset). In contrast, codes biased toward non-local task performance (squares) achieve total errors that are intermediate between oracle and static codes and that are minimized for an intermediate bias of $\alpha_{NL} = 0.61$. For display purposes, the error values in d–f were smoothed with a moving window of width 8 to average over sampling variations (Methods); see Extended Data Fig. 1 for unsmoothed results.

then used to adapt the encoding parameters $\psi = \{k, s_0\}$ for the task of accurately decoding the filter output. We measure the instantaneous decoding error as $e_t = (s_t - \hat{s}_t)^2$, and we average this error over time and context switches to compute the global decoding error.

We first construct a naively adaptive code that uses its context estimate to select among the local, context-optimal non-linearities

shown in Fig. 1c. The parameters $\{\psi_c^{\text{local}}\}$ of these non-linearities were chosen to minimize the error in decoding filter outputs sampled from each individual context c . We compare this code to an oracle code that uses the true context to select among these same non-linearities (black bar, Figs. 1d and 3b) and a static code that uses a single global non-linearity whose parameters ψ^{global} were

chosen to minimize the error in decoding filter outputs sampled from all contexts (white bar, Figs. 1d and 3b). The naively adaptive code achieves a global error intermediate between that of the static and oracle codes, a large fraction of which arises during periods of mismatch (Fig. 3b).

To reduce the effect of these periods of mismatch, we bias the naively adaptive code (Fig. 3c, left) toward inference (Fig. 3c, middle, with parameters $\{\psi_c^{\text{infer}, \alpha_1}\}$) or non-local task performance (Fig. 3c, right, with parameters $\{\psi_c^{\text{nonlocal}, \alpha_{\text{NL}}}\}$). In the former case, we bias the parameters of the non-linearity optimized for a particular context c toward the non-linearity that maximizes the probability $\sum_c p(\hat{c} = c)p(c)$ of correctly classifying all nine contexts on average and that thereby best discriminates between contexts. In the latter case, we bias the same parameters toward the non-linearity that minimizes the error in decoding filter outputs sampled from all other contexts $c' \neq c$; we sample these filter outputs in proportion to the probability of estimating the true context c' as the erroneous context c (that is, $s \sim \sum_{c'} p(s|c')p(\hat{c} = c|c')$; Methods). In both cases, biased parameters are constructed via a weighted combination of naively adaptive and fully biased parameters, with weights specified by α_1 and α_{NL} (that is, $\psi_c^{\text{infer}, \alpha_1} = (1 - \alpha_1)\psi_c^{\text{local}} + \alpha_1\psi_c^{\text{infer}}$ and $\psi_c^{\text{nonlocal}, \alpha_{\text{NL}}} = (1 - \alpha_{\text{NL}})\psi_c^{\text{local}} + \alpha_{\text{NL}}\psi_c^{\text{nonlocal}}$). Although these two different forms of biasing lead to seemingly subtle differences in the resulting non-linearities, the effect on overall performance is substantially different (Fig. 3d–f and Extended Data Fig. 1). Biasing toward inference by reducing the duration of mismatch leads to a substantial increase in the mismatched error rate (Fig. 3d, inset). This leads to higher mismatched error, which, together with the dramatic increase in the matched error (Fig. 3e, inset), ultimately results in higher global error (Fig. 3f, inset). In contrast, biasing toward non-local task performance by reducing the mismatched error rate leads to a weak reduction in the duration of mismatch, which thereby leads to a reduction in the mismatched error rate (Fig. 3d, square markers). This reduction is accompanied by an increase in matched error, and the global error is thus minimized for an intermediate bias (starred points in Fig. 3e–f). This highlights

how biasing away from the strategy that would be locally optimal if the context were stationary can improve global performance when the context is changing.

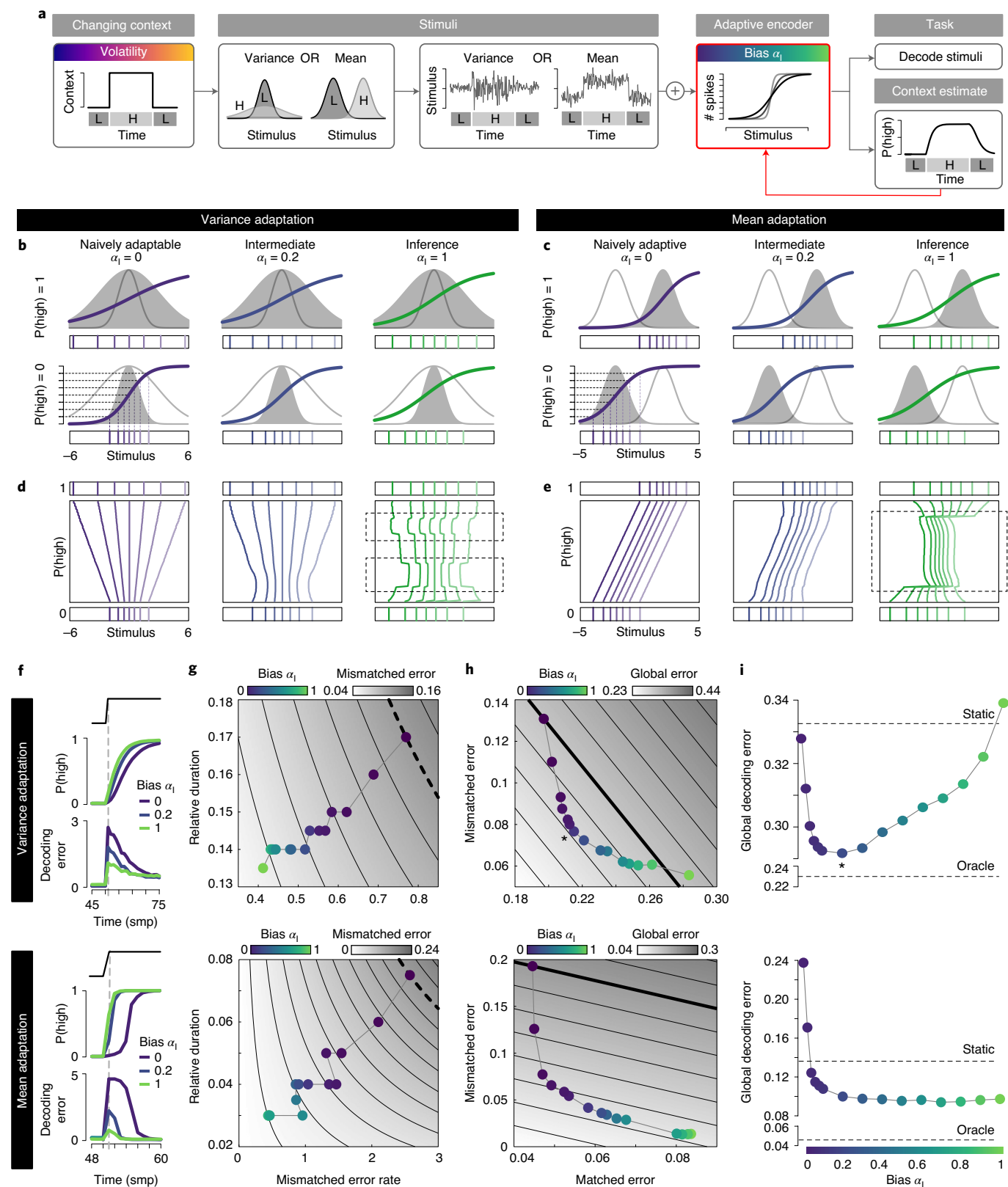
A normative explanation of the dynamics underlying variance and mean adaptation. We now consider a simplification of the previous scenario that mimics one of the most widely studied adaptation paradigms in sensory neuroscience—adaptation to changes in stimulus variance and mean^{2,4,8,9,31,32}. In this scenario, where stimulus statistics are highly controlled, the ideal observer model for inferring changes in these statistics is well known and provides a continuous estimate of the changing sensory context²¹. We construct a family of codes that use this estimate to optimally balance different sources of error (Fig. 4), and we show that these codes explain a diversity of adaptive dynamics that have been observed experimentally (Fig. 5).

Performance tradeoffs generalize across variations in stimulus statistics. To model this paradigm, we consider a simplification of the setup in Fig. 3a, shown in Fig. 4a, in which the environment switches between a ‘high’ and a ‘low’ context ($c = c_H$ and $c = c_L$, respectively). We separately vary the environmental volatility by adjusting the probability of these context switches. We take this changing latent context to specify either the variance (‘variance adaptation’) or the mean (‘mean adaptation’) of a Gaussian stimulus distribution (that is, $p(s_t|c_t) = \mathcal{N}(s_t; \mu, c_t^2)$ or $\mathcal{N}(s_t; c_t, \sigma^2)$, respectively). Stimuli from these distributions are then encoded by a simple model neuron. Specifically, at time t , a stimulus s_t is drawn from the distribution specified by the context c_t , transformed via a sigmoidal non-linearity with adaptable slope k and offset s_0 , distorted by noise, and, finally, discretized into one of n different response levels to produce a response r_t . These response levels can be interpreted as spike counts or discriminable firing rates, and their discrete nature limits the precision with which incoming stimuli can be encoded in the response r_t . This response is then linearly decoded to extract a stimulus estimate $\hat{s}_t$, which is used to update an

Fig. 4 | Biased codes improve performance across a range of stimulus conditions. **a**, We consider a simple model neuron in an environment that can switch between a low (L) and a high (H) context over time. This context signals either the variance or the mean of a Gaussian stimulus distribution. The neuron encodes incoming stimuli in a discrete response via a saturating non-linearity. This response is used to estimate the posterior belief $P(\text{high}) = P_t^H$ that the environment is in the high context at time t , which can be fed upstream (red arrow) and used to adapt the encoding non-linearity for the task of accurately decoding stimuli. We consider a family of codes that bias toward maintaining an accurate estimate of context (blue-green colormap), and we track performance in settings with varying volatility (purple-yellow colormap) (results in Extended Data Fig. 4). **b–e**, Optimal non-linearities vary as a function of the system's posterior belief $P(\text{high})$ about the current context, and they differ among the naively adaptive (purple, left), fully biased (green, right), and intermediate (blue, center) codes. To visualize these changes, we use vertical lines to differentiate sets of stimuli that are encoded in different response levels (illustrated in the lower-left panels of **b** and **c**); regions where these lines are closer together mark steeper portions of the non-linearity. **b, d**, When the naively adaptive code (left) is certain of either the high- or low-variance context ($P(\text{high}) = 1$ or 0), the non-linearity is either broad or narrow, respectively (**b**; upper and lower left), and linearly scales between these extremes (**d**, left). When the fully biased code (right) is more certain of the high-variance context ($0.5 < P(\text{high}) < 1$), the non-linearity broadens and shifts toward the tail of the high-variance stimulus distribution (**b**, upper right; **d**, right, upper dashed box; note that we observe two symmetric solutions shifted toward either tail (Extended Data Fig. 3a); this solution produces fewer spike counts on average). When the code is more certain of the low-variance context ($0 < P(\text{high}) < 0.5$), the non-linearity sharpens and centers (**b**, lower right; **d**, right, lower dashed box). The optimal code (**b** and **d**, center) shares features of the naively adaptive and fully biased codes. **c, e**, When the naively adaptive code (left) is certain of either the high- or low-mean context ($P(\text{high})$ near 1 or 0), the non-linearity is centered on the high- and low-mean distributions, respectively (**c**, upper and lower left) and linearly scales between these extremes (**e**, left). When the fully biased code (right) is certain of either context, the non-linearity is broad and shifted toward the distribution that signals the other context (**c**, right); when the code is uncertain, the non-linearity is sharp and centered between the two distributions (**e**, dashed box). An intermediate code (**c** and **e**, center) shares features of the naively adaptive and fully biased codes. The abrupt changes in the non-linearities shown in the rightmost panels of **d** and **e** arise from bifurcations in the error landscape for small variations in $P(\text{high})$ (Extended Data Fig. 3). **f**, After an abrupt increase in variance (upper panel) or mean (lower panel), the system updates its posterior belief $P(\text{high})$, which is accompanied by a transient increase in decoding error over time (measured in samples (smp)). **g**, Biasing the code toward inference leads to shorter periods of mismatch and lower mismatched error rates during both variance adaptation (upper panel) and mean adaptation (lower panel). Solid lines, contours of constant mismatched error. **h**, Biased codes exhibit a tradeoff between matched and mismatched error during both variance adaptation (upper panel) and mean adaptation (lower panel). Solid lines, contours of constant global error. **i**, During variance adaptation (upper panel), most biased codes outperform the naively adaptive and static codes, and optimal performance is achieved for an intermediate bias of $\alpha_i = 0.2$ (star). During mean adaptation (lower panel), all biased codes outperform the naively adaptive code, and performance monotonically improves with increasing bias.

internal estimate of the current context $\hat{c}_t$. To construct this estimate, we use an optimal Bayesian observer that is identical in form to that derived in ref. ²¹ but uses decoded stimulus estimates rather than raw stimulus values to update its context estimate. This observer first computes the posterior probability $P_t^H = p(c_t = c_H | \hat{s}_{t \leq t})$ that the environment is in the high context at

time t and then averages this posterior over contexts to produce a continuous context estimate $\hat{c}_t$. This context estimate is then used to adapt the encoding parameters $\psi = \{k, s_0\}$ for the task of accurately decoding the stimulus. As before, we average the instantaneous decoding error, $e_t = (s_t - \hat{s}_t)^2$, over time and context switches to compute the global decoding error.



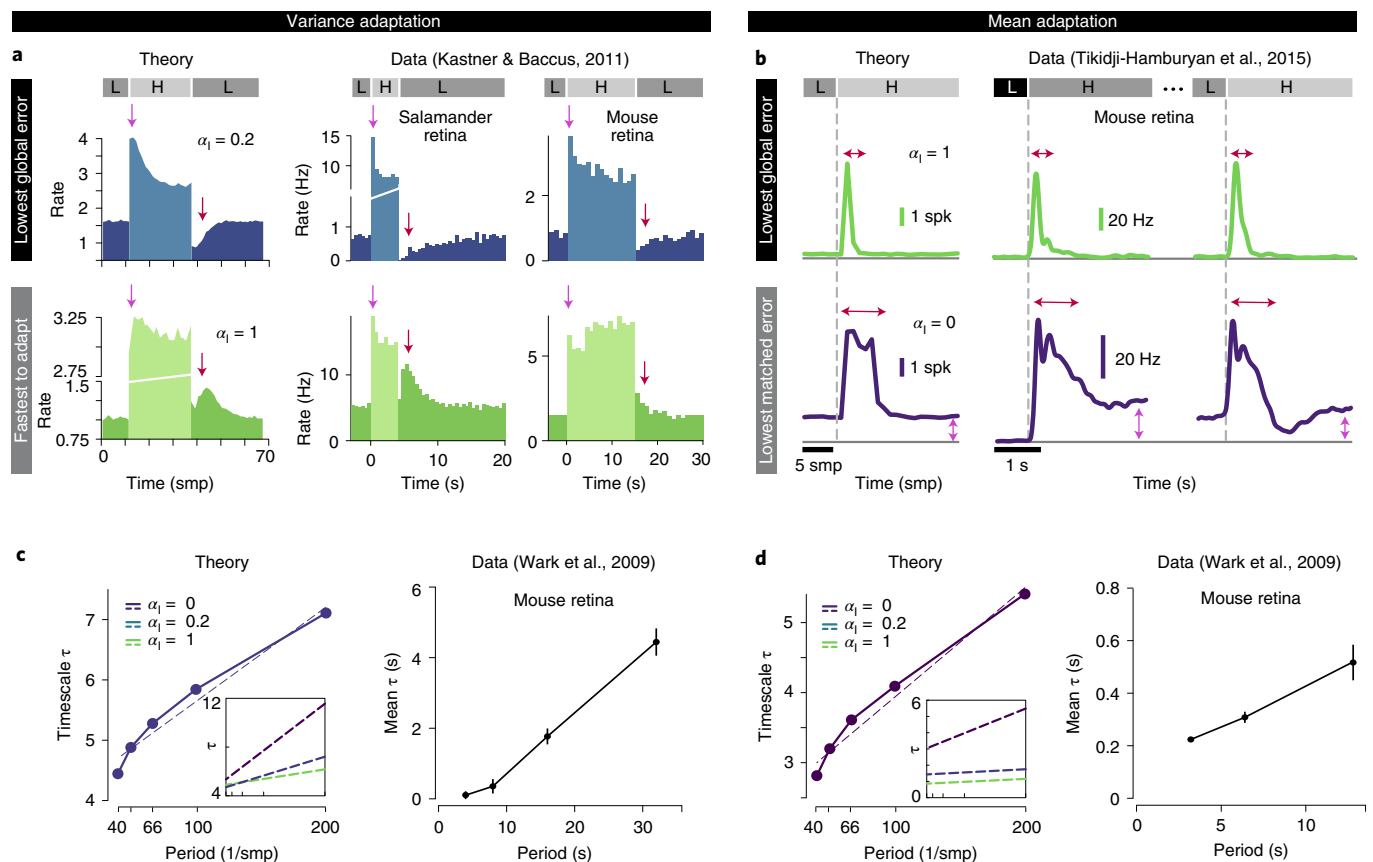


Fig. 5 | Biased codes capture diverse profiles and timescales of adaptive dynamics. **a, b**, Average firing rates change in response to a step increase and decrease in variance (**a**) and mean (**b**). **a**, The code that minimizes global error (upper left, $\alpha_1 = 0.2$) produces a transient drop in firing rate after a change from high (H) to low (L) variance (red arrow); in contrast, the code that minimizes adaptation time (lower left, $\alpha_1 = 1$) produces a transient increase in firing rate (red arrow). Both codes produce transient increases in firing rate after a change from low to high variance (pink arrows). Similar dynamics have been observed in ‘adapting’ (upper row, right panels) and ‘sensitizing’ (lower row, right panels) ganglion cells in the retina of salamander (middle column) and mouse (right column) in response to step changes in contrast (figure reproduced from, ref. ², Nature Publishing Group). **b**, The code that minimizes global error and simultaneously minimizes adaptation time (upper left, $\alpha_1 = 1$) produces a sharp and transient increase in firing rate after a change from low (L) to high (H) mean (red arrow). In contrast, the code that minimizes matched error (lower left, $\alpha_1 = 0$) produces a higher baseline firing rate (pink arrow), as well as a longer and doubly peaked transient increases after a change in mean (red arrow). Similar dynamics have been observed in the mouse retina, shown for two ON ganglion cells (different rows in right panels) responding to a step increase in luminance at lower (middle column) and higher (right column) light levels (figure reproduced from, ref. ³⁵, Nature Publishing Group). **c, d**, The timescale of adaptation to changes in variance (**c**, left) or mean (**d**, left) increases with the period of these changes. This relationship is well approximated by a linear fit (dashed line) whose slope and offset varies as a function of the bias α_1 (inset). In mouse retinal ganglion cells (**c** and **d**, right), the timescale of adaptation exhibits similar scaling with the switching period of the stimulus distribution (figure reproduced from ref. ³², Cell Press). Timescales were estimated using an exponential fit (Methods).

We first construct a naively adaptive code whose encoding parameters are chosen to accurately decode stimuli given the system’s current context estimate. Because this estimate is continuous, we construct a set of encoding parameters that take on a continuum of values. To this end, the parameters $\{\psi_{\hat{c}}^{\text{local}}\}$ are chosen to minimize the expected error $\langle (s - \hat{s})^2 \rangle_{p(s|\hat{c})}$ in decoding stimuli sampled from the system’s estimate of the stimulus distribution, $p(s|\hat{c}_t)$. We then bias this code toward inference by determining the parameters $\{\psi_{\hat{c}}^{\text{infer}, \alpha_1}\}$ that minimize a weighted combination of this expected decoding error and the expected error in estimating the context, $\langle (\hat{c}(s) - \hat{c}(\hat{s}))^2 \rangle_{p(s|\hat{c})}$, where α_1 specifies the weight between these two error terms (see Extended Data Fig. 2 for a tractable approximation using linear interpolation). We empirically found that this strategy, rather than one biased toward non-local task performance, produced the largest reduction in global error; we therefore focus on this strategy for the remainder of this section. We compare the performance of these biased codes to a static code whose parameters $\psi_{\hat{c}}^{\text{global}}$ are chosen to minimize the average error in decoding stimuli

sampled from both contexts and an oracle code whose parameters $\{\psi_{c_L}^{\text{local}}, \psi_{c_H}^{\text{local}}\}$ are chosen to minimize the average error in decoding stimuli sampled from either the low or high contexts, respectively.

Naively adaptive and biased codes differ in how the neuron’s limited response levels are allocated to different regions of the stimulus space given the observer’s posterior belief about the underlying context (Fig. 4b–e, left). The naively adaptive code uses a strategy that directly tracks this belief by scaling the width of the non-linearity when the variance is changing in time (Fig. 4b,d, left) and shifting the location of the non-linearity when the mean is changing in time (Fig. 4c,e, left). In contrast, a fully biased code implements a form of uncertainty-dependent change detection (Fig. 4b–e, right columns, and Extended Data Fig. 3). When the system is certain of either context, the fully biased code shifts the non-linearity toward the alternate context, thereby hedging its bets toward the change that is likely to occur. When the system is uncertain, it adjusts the non-linearity to resolve this uncertainty. This manifests in different behavior during variance versus mean adaptation. In the former case,

increases in variance are easier to detect than decreases²¹, and, thus, the behavior differs depending on whether the system is more certain of the high versus low context. When more certain of the high context, the fully biased code uses a broad and shifted non-linearity to resolve stimuli of large magnitude that would signal high variances (upper dashed box in Fig. 4d, right). When more certain of the low context, it uses a sharp and centered non-linearity to resolve stimuli of small magnitude that would signal low variances (lower dashed box in Fig. 4d, right). In the latter case, when the mean is changing, the behavior is symmetric to variations in uncertainty. When the system is sufficiently uncertain of either the high or low context, the fully biased code uses a narrow non-linearity centered between the two stimulus distributions to accurately resolve stimuli that would differentiate high versus low means (dashed box in Fig. 4e, right).

The non-linearities in Fig. 4b–e show the optimal partitioning of the stimulus space as a function of the system's belief about the current context. In a dynamic setting, this belief will be continually changing, as shown in Fig. 4f for a switch in variance (upper panel) and mean (lower panel). In both cases, the naively adaptive code is the slowest to update its belief, and it produces the largest transient errors during this time. Biased codes reduce both this duration of mismatch and the corresponding error rate (Fig. 4f,g). This leads to an overall reduction in mismatched error but an increase in matched error, and, thus, the global error is minimized for a non-zero bias (starred points in Fig. 4h,i). We observe qualitatively similar results as we vary the volatility of context switches, but we find that more volatile environments require larger biases to minimize global error (Extended Data Fig. 4).

Different balances of matched and mismatched error explain a diversity of adaptive neural dynamics. The improvements in performance shown in Fig. 4 are accompanied by dynamic changes in the firing rate of our model neuron. To extract this firing rate, we used an entropy coding scheme to recode the output of the neuron's non-linearity (Methods and Extended Data Fig. 5). The resulting firing rates exhibit a range of different patterns that resemble a diversity of adaptation dynamics observed in salamander, mouse, rabbit, and primate retina (Fig. 5a,b; refs. 2,33–35).

During variance adaptation, codes that minimize either global error or adaptation time exhibit dynamics resembling those of so-called 'adapting' and 'sensitizing' cells, respectively (Fig. 5a; refs. 2,33,34). During mean adaptation, codes that minimize either global error (and simultaneously adaptation time) or matched error exhibit dynamics that resemble different ganglion cells in mouse retina (Fig. 5b; ref. 35). Within our model, this diversity of dynamics is realized by varying a single parameter α_1 that controls the balance of distinct error terms that arise when coding in non-stationary environments. The observed match to a variety of different neural responses thus suggests that the corresponding cell types are optimized to balance these sources of error to maintain high performance in the face of changing stimulus statistics.

In all cases, the speed of adaptation depends on the rate of context switches. When this rate is low, the system expects the environment to be more stable and thus requires more stimulus samples to be convinced of a change in context. This leads to slower changes in the posterior belief (Fig. 4f and Extended Data Fig. 6) and slower timescales of adaptation (Fig. 5c,d, left panels), with smaller biases leading to slower overall timescales (insets of Fig. 5c,d). Similar relationships have been observed in the fly, mouse, rat, and electric fish (Fig. 5c,d, right panels; refs. 7,8,32,36). Although existing models can separately explain either this relationship between adaptation speed and environmental volatility³² or the dynamics of adapting and sensitizing cells², no existing framework can simultaneously account for—and provide a unified normative explanation of—both sets of phenomena.

Tradeoffs in receptive field adaptation to higher-order stimulus statistics. To illustrate the generality of this framework beyond single neurons and low-order stimulus statistics, we consider a simple feed-forward network of two neurons that uses an adaptive set of linear receptive fields to jointly encode local patches of natural images (Fig. 6a). To maximize information transmission, these receptive fields must be adapted to the local statistical structure of these patches³⁷, which depends strongly on their location within a visual scene³⁸. Here, we use unsupervised learning to extract four statistical contexts that roughly correspond to dominant cardinal orientations and delineate different spatial regions of visual scenes (Fig. 6b and Methods).

As before, we consider an environment that can dynamically switch among these four contexts. At each time t , an image patch s_t is sampled from the context specified by c_t , encoded via the receptive fields of the two neurons in the joint response $\mathbf{r}_t$, and used to update an ML estimate $\hat{c}_t$ of the current context. This estimate is then used to adapt the receptive fields, specified by the encoding parameters ψ_t , for the task of maximizing information transmission over a window of time ΔT . In the low-noise limit, information transmission will be high when each neuron saturates the variance of its output, and neural activity is decorrelated^{37,39}. We can thus measure the task error at time t using the symmetrized Kullback–Leibler (KL) divergence between this idealized and the actual output distributions, computed using neural activity over the past ΔT time points (Fig. 6c and Methods). We then average this error over time and context switches to compute the global KL error.

The set of context-optimal receptive fields $\{\psi_c^{\text{local}}\}$ that maximize information about patches within each context c differ from one another (Fig. 6d, lower rows) and from the set of global receptive fields ψ^{global} that maximize information about patches uniformly sampled from all contexts (Fig. 6d, upper row). Such adaptation has been observed in primary visual cortex but typically evokes more subtle effects than those generated by this simplified model^{10,40,41}. We construct a naively adaptive code that uses its context estimate to select among the context-optimal receptive fields. We then bias this code toward non-local task performance by learning a set of receptive fields $\{\psi_c^{\text{nonlocal}, \alpha_{\text{NL}}}\}$ that maximize information about a set of image patches in which a fraction $(1 - \alpha_{\text{NL}})$ is sampled from the current context, and a fraction α_{NL} is sampled from the remaining contexts. We empirically found this biasing strategy to yield the largest reduction in global error.

As in our previous examples, stronger biases lead to lower mismatched error but higher matched error (Fig. 6e). As a result, the optimal performance that minimizes global error is achieved for an intermediate bias (starred points in Fig. 6e–g). Interestingly, subtle modifications of receptive field structure lead to a dramatic decrease in the duration of mismatch (Fig. 6f) at the cost of a slight increase in global error (Fig. 6g), a tradeoff that could be exploited by a neural population that balances adaptation speed and information transmission. Taken together, these results illustrate the possible benefits of biasing the joint encoding of high-dimensional stimuli to improve global performance.

Discussion

Adaptive phenomena in neural coding have been a subject of experimental and theoretical research since Adrian's and Zotterman's discovery of adaptation¹. The dynamics of adaptation have traditionally been characterized by changes in experimentally observable quantities—such as firing rates, tuning curves, and receptive fields—in response to changes in the statistics of incoming sensory stimuli^{4,7,17,42,43}. A large body of modeling work has focused on reproducing these dynamic changes^{36,44} or on simulating their mechanistic origins^{2,33,45}. Other studies, both theoretical^{46,47} and experimental⁵, have examined the effect of adaptation dynamics on neural coding and perception. Together, these approaches can

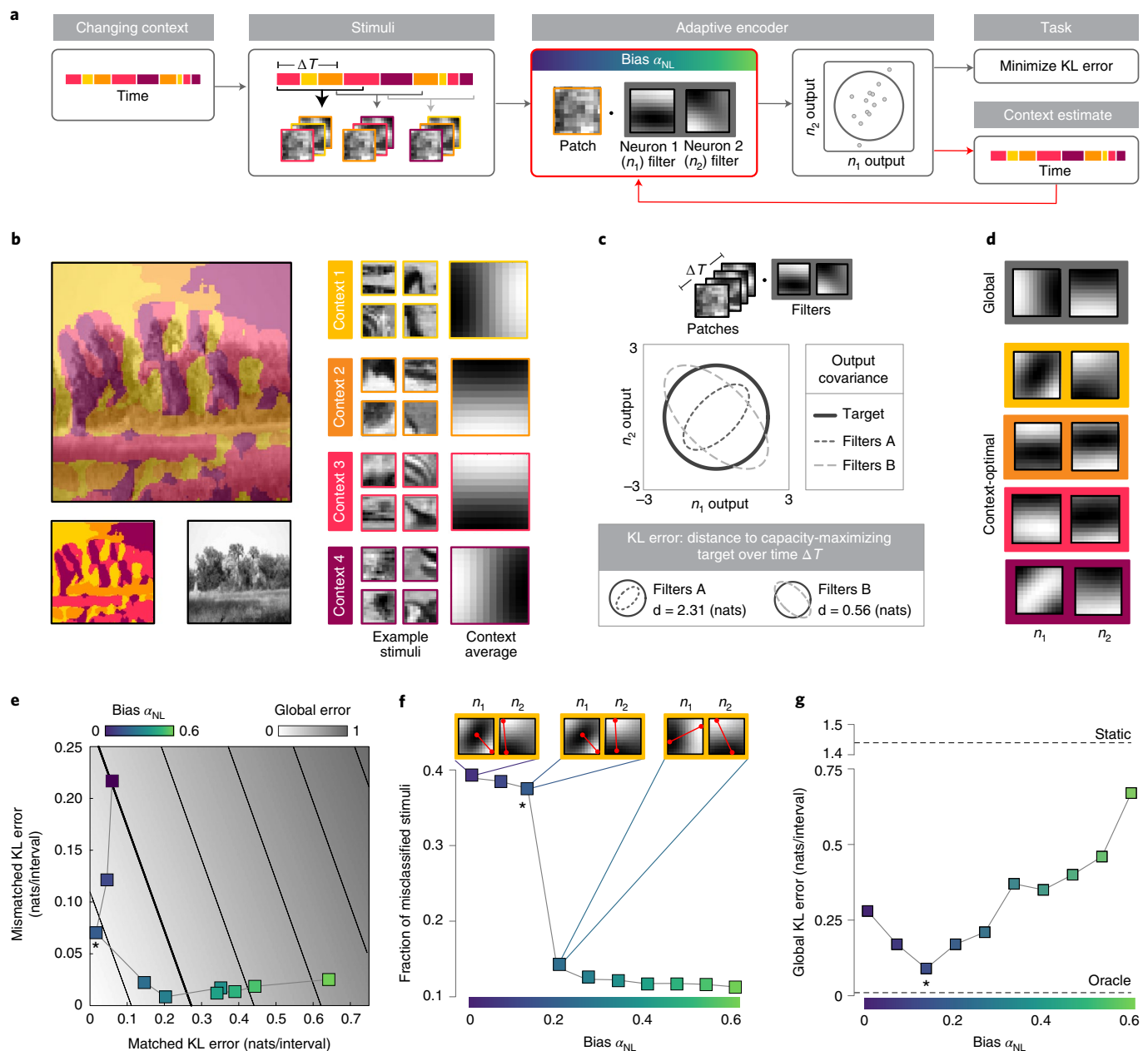


Fig. 6 | Feed-forward network can increase information transmission by biasing away from local stimulus statistics. a, We consider a simple feed-forward network of two visual neurons that uses linear receptive fields to encode incoming image patches. These patches originate from different contexts that can switch over time. The joint response of these neurons is used to construct an estimate of the current context, which is then fed upstream (red arrow) and used to adapt the receptive fields for the task of maximizing information transmission. **b**, Contexts with similar statistical structure correspond to different regions of a visual scene (left column). We consider four contexts that were discovered by unsupervised learning and that roughly correspond to the orientation of features within individual image patches (middle column; Methods); the average image patch within each context summarizes the differences between contexts (right column). **c**, To quantify the error in information transmission during a time interval ΔT (schematic in upper row), we estimate the symmetrized KL divergence (KL error, measured in natural units (nats)) between the approximate distribution of network outputs over this interval (dashed gray ellipsoids) and the target output distribution that maximizes information transmission (solid black circle). **d**, Information-maximizing filters are learned with ICA³⁷. Global filters that maximize information transmission for all image patches (gray frame) differ from context-optimal filters (colored frames) that maximize information transmission for patches originating from individual contexts. **e**, Biasing the code toward reducing the mismatched error rate yields a tradeoff between mismatched and matched error. The optimal code that minimizes global error is marked with a star. Solid lines: contours of constant global error. **f**, Biased codes reduce the duration of mismatch, measured via the classification error per stimulus sample. Increasing the bias beyond the optimum results in a dramatic improvement of classification performance (main panel). This is accompanied by an abrupt change in the shape of receptive field (upper row, shown for Context 1). Red barbells mark the maximum and minimum values of the receptive fields and are shown purely for visualization. **g**, Biased codes achieve performance intermediate between the static and oracle codes. The optimal code ($\alpha_{NL} = 0.14$, star) achieves three-fold lower error than the naively adaptive code ($\alpha_{NL} = 0$).

capture the dynamics of adaptation and explain their effect on specific computations. However, they do not specify the computations that would warrant such dynamics, nor can they explain why a particular set of dynamics might be more or less beneficial to an organism. Such questions fall within the domain of normative approaches to neural coding.

Existing normative accounts of sensory adaptation have largely been developed within the framework of the efficient coding hypothesis¹⁸. Efficient coding studies traditionally derive optimal solutions for maximizing utility^{12,24,37} or minimizing error^{20,29,48} in representing natural sensory stimuli under the assumption that their distribution is stable^{12,28}. Natural environments, however, are constantly changing^{17,38}, which is reflected in the dynamic and adaptive structure of sensory codes in the brain^{8,22,49}. To account for this, efficient coding posits that adaptation readjusts the sensory code to maintain a maximal rate of information transmission^{8–10,17}, supporting the idea that adaptation reflects a dynamic reallocation of limited metabolic resources that serves to maximize performance within different environmental contexts. Although these studies provide a normative objective for the outcome of adaptation, they do not explain the dynamic structure of resource reallocation that would best achieve this outcome, nor do they explain how this reallocation should optimally unfold in non-stationary environments.

In this study, we addressed this gap by extending classical theories of efficient coding to the study of adaptive neural computation in non-stationary environments. Our key insight is that efficient encoding of task-relevant information can require different solutions when the statistics of the environment are changing than when they are stationary. In such scenarios, optimal sensory systems need to not only accurately encode incoming stimuli but also detect and adapt to changes in their underlying distribution⁹. To this end, it has previously been suggested that adaptation is guided by an internal model that accumulates information over time to infer latent properties of the stimulus distribution^{21,32}, and several lines of work have shown that relative timescales of adaptation are consistent with predictions of ideal observer models^{7,21,33}. However, such online estimation of changing stimulus statistics is further complicated if the encoding of these stimuli is lossy⁵⁰. As a result, there will necessarily be periods of time when the system incorrectly estimates the stimulus distribution, and the code is maladapted. Such periods of mismatch will result in suboptimal use of resources—as has been observed experimentally via a drop in information rate immediately after a change in stimulus statistics^{8,9}—and, consequently, larger coding errors than would be achieved during periods of stationarity. To combat these adversarial effects of mismatches, we formalized the relationship between a resource-constrained encoder and an ideal observer, and we identified coding strategies that could optimally exploit the observer's context knowledge to improve coding performance. We highlighted cases where the optimal code is not the one designed to maximize task performance given a local context estimate but, rather, the one designed to simultaneously balance errors that arise when this estimate is inaccurate. We then showed how this general theoretical setting can be used to derive quantitative predictions about neural dynamics that can be compared against phenomenological and mechanistic models of adaptation^{3,36,44}.

This framework enables comparisons across different coding objectives. At one extreme, we identified codes that favor fine discrimination of stimuli within one context at the cost of resolving stimuli that would signal changes in context. At the other extreme, we identified codes that favor detecting changes over making fine discriminations. Previous work showed that adaptation can mediate tradeoffs between detection and discrimination at both a neural and behavioral level⁵. In this study, we showed how such a shift should evolve over time, how it might be signaled by specific patterns of neural dynamics, and how these two different objectives might optimally be balanced. Because we used decoding error to

evaluate performance, our optimal solutions ultimately favor local discrimination and use detection only as a means of improving discrimination. Other systems might instead favor the detection of context changes over the discrimination of local stimulus details⁵⁰. Moreover, different tasks might warrant additional performance criteria; for example, tasks such as predator localization or visual speed estimation might necessitate codes that maintain errors below a critical threshold rather than minimize global error. Although these different objectives are handled separately within our current framework, it might be possible to derive coding schemes that optimize them simultaneously. The question of how such criteria should be balanced within a population, and how these objectives might relate to different functional cell types, is the subject of future work.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41593-021-00846-0>.

Received: 12 June 2019; Accepted: 24 March 2021;

Published online: 20 May 2021

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Methods

Local contrast and luminance adaptation in a linear–non-linear neuron.

Adaptive encoder. We modeled a classical linear–non-linear Poisson neuron that encodes incoming image patches in neural responses. This model has been broadly used to model stimulus response properties in a range of sensory systems²⁵. Here, this model neuron encodes incoming image patches of size 12×12 pixels by projecting them onto an oriented linear filter ϕ . The filter was randomly selected from a set of 64 filters learned with logistic independent component analysis (ICA)³⁷ from a dataset of images consisting of natural movie frames used in ref.⁵¹ (we used images from the UPenn Natural Image Database⁵² as illustrations in Figs. 1, 2, and 6). The specific filter that we used is shown in Fig. 1a; the choice of this filter did not qualitatively affect the results described here. The output of the filter s_t was mapped onto firing rate λ_t via an adaptive logistic non-linearity:

$$\lambda_t = \frac{\lambda_{\max}}{1 + \exp(-k(s_t + \xi_t - s_0))} \quad (4)$$

where $\lambda_{\max}$ is the maximal firing rate; $\psi = [k, s_0]$ are adaptive encoding parameters that specify the slope and offset of the non-linearity, respectively; and $\xi_t \sim \mathcal{N}(0, \sigma_\xi^2)$ is an additive Gaussian noise with variance $\sigma_\xi^2 = 0.01$. The discrete spike count response of the neuron r_t was drawn from a Poisson distribution parameterized by rate $\lambda_t \Delta T$. For simplicity, we assume the time interval of a response to be equal to 1 (that is, $\Delta T = 1$).

Model environment. The encoder adapts by modifying the parameters of the encoding non-linearity given a fixed filter ϕ . We therefore constructed contexts that capture the variability in its inputs (that is, in the filter outputs) observed across an ensemble of natural scenes. We considered an environment that could switch between nine such contexts at a small but fixed probability per unit time, p_s (that is, $p(c_t | c_{t-1}) = p_s/8$ if $c_t \neq c_{t-1}$ and $(1 - p_s)$ otherwise). All results were generated with $p_s = 0.01$.

To mimic the variability in filter outputs that would be generated by visual exploration of a scene, we simulated saccadic trajectories across different regions of natural images. For a given trajectory, the distribution of filter outputs was well approximated by a Gaussian distribution; we, thus, used the set of trajectories to define the means and variances of a set of Gaussian contexts that spanned the range of filter outputs generated during this exploration. To generate the trajectories, we used the same dataset of images described above, and we standardized each image in this dataset to have zero mean and unit variance. Each trajectory was generated via a Gaussian random walk and was used to extract a sequence of 10^3 image patches of size 12×12 pixels; that is, at each time step, the center of the image patch was moved by n_x and n_y pixels in the horizontal and vertical directions, respectively, where n_x, n_y were drawn from a Gaussian distribution with zero mean and unit variance and rounded to the nearest integer. We randomly initialized the starting location of each trajectory, and we simulated 5×10^4 trajectories in total. Three such trajectories are shown in the upper panel of Fig. 1b. We then filtered the image patches within each trajectory using the oriented filter ϕ , which produced a single distribution of filter outputs for each trajectory (illustrated in the lower panels of Fig. 1b for the corresponding trajectories). We then empirically estimated the mean and variance of each trajectory-specific distribution of filter outputs, thereby generating a distribution of means and variances across trajectories. We then selected three variance values σ_i^2 (0.6, 2.9, 13.6) that logarithmically spanned the range between the 10th and 90th percentiles of the distribution of variances and three mean values μ_j (−2.3, 0, 2.4) that linearly spanned the range between the 1st and 99th percentiles of the distribution of means. We then used all $3 \times 3 = 9$ combinations of means and variances to define a set of contexts $c = \{1, \dots, 9\}$, where each of the context labels specifies a pair of variance and mean indices i, j . Each sensory context, thus, defines a Gaussian distribution of filter outputs with variance σ_i^2 and mean μ_j , such that $p(s_t | c) = \mathcal{N}(\mu_j, \sigma_i^2)$. These distributions are shown in Fig. 1c.

Sensory task. We evaluated the performance of the system by measuring the average error in decoding stimuli, $\langle (s - \hat{s})^2 \rangle$. To compute this error, we decoded an estimate of the filter outputs using ML decoding. At a given time t , we took the ML estimate $\hat{s}_t$ to be the value that maximizes the probability of observing the response r_t given the encoding parameters k, s_0 :

$$\hat{s}_t = \arg \max_{s_t} p(r_t | k, s_0, s_t) \quad (5)$$

To numerically simplify the decoding process, we discretized the range of decoded filter outputs into 2^{14} values uniformly spaced between −16 and 16.

Observer. To aid in task performance, the system constructs an internal estimate of context $\hat{c}_t$ that is used to select the appropriate encoding parameters at each time step. We used an ML estimation scheme to infer the context identity $\hat{c}$ from the neural response r_t . For a given nonlinearity parametrized by its slope k and offset s_0 , and for a given context c parametrized by mean μ_j and variance σ_i^2 , we empirically

estimated the distribution of spike counts $p(r_t | k, s_0, c)$. Over a window of time ΔT , the log-likelihood of responses r_t is given by:

$$\mathcal{L}(c) = \log p(r_{t-1}, \dots, r_{t-\Delta T+1} | k, s_0, c) \propto \sum_{\tau=0}^{\Delta T-1} \log p(r_{t-\tau} | k, s_0, c), \quad (6)$$

where $p(r_t | k, s_0, c) = \int p(r_t | k, s_0, s_t) p(s_t | c) ds_t$ is the distribution of neural responses in context c , given non-linearity parameters k, s_0 . We note that, at times, there might be a mismatch between the true context and the estimated context, which results in an incorrect likelihood function. The ML estimate of the current context $\hat{c}$ was found by iterating over all context labels c and returning those that maximize the log-likelihood of the past ΔT responses:

$$\hat{c} = \arg \max_c \mathcal{L}(c) \quad (7)$$

We set the history length ΔT to 20 steps, which corresponds to one-fifth of the average duration of each context given the switching probability $p_s = 0.01$. For notational simplicity, we refer to $\hat{c}$ as the context estimate, and we will write $\hat{c} = c$ if the context estimate was correct (that is, equal to the true context c) and $\hat{c} \neq c$ if the context estimate was incorrect.

Optimization of non-linear response functions. The set of encoding parameters $\mathcal{P} = \{\psi\}$ was optimized offline for each context c . During online simulations (described below), a single set of encoding parameters was selected from this set at each time step t given the system's estimate of context $\hat{c}_t$.

We first determined the encoding parameters $\psi_c^{\text{local}} = \{k, s_0\}_c$ that minimize decoding error in each context c ; these parameters specify the ‘context-optimal’ non-linearities shown in Fig. 1c. We performed an exhaustive grid search over a 32×32 grid of parameter values, spanning the ranges $[0.5, 1]$ for the slope k and $[-4, 4]$ for the offset s_0 . We generated 2×10^4 stimulus samples s from the Gaussian distribution of filter outputs for each context. We then passed this set of stimulus samples through the encoding non-linearity, defined by a given combination of slope k and offset s_0 , to generate the corresponding firing rates λ . We then randomly sampled a spike count r from a Poisson distribution parameterized by the rates λ . We used the ML decoding procedure described in the subsection ‘Sensory task’ above to decode each stimulus and thereby obtain stimulus estimates $\hat{s}$. For each parameter combination, we computed the decoding error $\langle (s - \hat{s})^2 \rangle_{p(s|c)}$ averaged over a given context-specific distribution of stimuli. For each context c , we selected the combination of slope k^c and offset s_0^c that minimized this average error.

An analogous procedure was used to determine the parameters ψ^{global} of the static non-linearity shown in Fig. 1c. Here, we uniformly sampled 2×10^4 stimulus values from all nine contexts. The same grid search was used to determine the combination of slope and offset that minimize the decoding error $\langle (s - \hat{s})^2 \rangle_{p(s)}$ averaged over the distribution of stimuli marginalized over contexts (that is, $p(s) = \sum_{c=1}^9 p(s|c)p(c)$, where $p(c) = 1/9$).

To reduce either the mismatched error rate or the duration of mismatch, we biased each context-optimal non-linearity toward a target non-linearity. We did this by constructing convex combinations of context-optimal parameters k^c, s_0^c and fully biased target parameters $k^{\text{B},c}, s_0^{\text{B},c}$, where $\text{B} \in \{\text{I}, \text{NL}\}$ denotes whether the target parameters are optimized for inference or non-local task performance. We weighted these combinations by the coefficient $\alpha_{\text{B}} \in [0, 1]$. For $m \in \{k, s_0\}$, we took $m^{\text{a},c} = (1 - \alpha_{\text{B}})m^c + \alpha_{\text{B}}m^{\text{B},c}$. We used 50 values of α_{B} spaced evenly between 0 and 1 to construct two families of codes that were biased toward minimizing either the error rate or the duration of mismatch. This biasing strategy approximates the set of non-linearity parameters by linearly interpolating between k^c, s_0^c and target parameters $k^{\text{B},c}, s_0^{\text{B},c}$. This approximative strategy, although not strictly optimal, results in a global error minimum that outperforms the naively adaptive code. Moreover, in a simplified model of mean and variance adaptation (Fig. 4), we found that a more principled, but more computationally exhaustive, biasing strategy can be well approximated by linear interpolation (Extended Data Fig. 2).

To bias codes toward reducing the duration of mismatch, we optimized a single target non-linearity for context inference (shown in Fig. 3c, right, $\alpha_{\text{I}} = 1$). We again performed an exhaustive grid search over a 32×32 grid of parameter values, here spanning the ranges $[0.1, 3]$ for the slope k and $[-4, 4]$ for the offset s_0 . We sampled 10^4 stimulus values from each context, resulting in 9×10^4 samples in total. We then generated corresponding firing rates, responses, and stimulus estimates as described above. For each individual neural response r_t we then used the ML decoding procedure described above to construct an estimate of context $\hat{c}$; note that this is equivalent to maximizing the log-likelihood in Equation (6) using a window of size $\Delta T = 1$. We then selected the pair of non-linearity parameters $k^{\text{B}}, s_0^{\text{B}}$ that maximized the probability $\sum_c p(\hat{c} = c | k, s_0) p(c)$ of correctly classifying all nine contexts on average. We note that a non-linearity optimized in this way can be used to infer the context using a time window ΔT larger than 1 (that is, using $(r_t, \dots, r_{t-\Delta T+1})$), following the i.i.d. assumption: $\mathcal{L}(c | (r_t, \dots, r_{t-\Delta T+1}), k^{\text{B}}, s_0^{\text{B}}) = \prod_{\tau=0}^{\Delta T-1} p(r_{t-\tau} | k^{\text{B}}, s_0^{\text{B}}, c)$.

To bias codes toward reducing the mismatched error rate, we constructed a different target non-linearity for each context (an example of which is shown in Fig. 3c, middle, $\alpha_{\text{NL}} = 1$). To minimize the mismatched error rate, we minimized

the decoding error averaged across the distribution of stimuli encountered by the system at times when the system's context estimate $\hat{c}$ differs from the true context c' (that is, $c' \neq \hat{c}$). We denote this distribution as $p_{\text{mis}}(s|\hat{c}) = \sum_{c' \neq \hat{c}} p(s|c')p(c'|\hat{c})$.

The distribution $p(c'|\hat{c})$ specifies the probability of mistaking the true context c' for the estimated context $\hat{c} \neq c'$. This distribution depends on the current estimate $\hat{c}$, because some contexts are easier to mistake for the current context than others. The pattern of these mistakes is summarized by the confusion matrix $M(\hat{c}, c) = p(c'|\hat{c})$, where $c' \neq \hat{c}$, can be obtained from the confusion matrix M by removing diagonal entries $p(\hat{c}, c = \hat{c})$ and renormalizing each row, such that $\sum_{c' \neq \hat{c}} p(c'|\hat{c}) = 1$.

We constructed a training dataset consistent with the distribution $p_{\text{mis}}(s|\hat{c})$ for each context $\hat{c}$ by running an online simulation (see below) with a naively adaptive code for 3×10^5 time steps. For each context, we then pooled together all stimulus samples that were incorrectly classified by the ML inference procedure (that is, stimulus samples s_i for which the resulting context estimate $\hat{c}_i$ was incorrect). We then performed an exhaustive grid search over a 32×32 grid of parameter values, here spanning the ranges $[0.5, 1]$ for the slope k and $[-4, 4]$ for the offset s_0 . We then generated corresponding firing rates, responses, and stimulus estimates as described above, and we selected the pair of non-linearity parameters that minimized the decoding error $\langle (s - \hat{s})^2 \rangle_{p_{\text{mis}}(s|\hat{c})}$.

Online simulations. We simulated dynamics over 3×10^5 time steps. At each time step t , the system maintained a context estimate $\hat{c}_{t-1}$ from the previous time step. This estimate was used to select the set of encoding parameters $\psi_t = \{k, s_0\}$ to be used on the current time step. We then randomly sampled a single stimulus sample s_i from $p(s|c_t)$. This stimulus sample was encoded in a response r_t , which was then used to update the context estimate $\hat{c}_t$. The response was also decoded to construct an estimate $\hat{s}_t$, which was then used to compute the instantaneous decoding error $(s_t - \hat{s}_t)^2$.

To define periods of match and mismatch, we selected all time points for which $\hat{c}_t = c_t$ and $\hat{c}_t \neq c_t$, respectively. The relative duration of mismatch, shown in Fig. 3d and in the inset of Fig. 3c, was computed by scaling the total number of mismatched time points by the total number of all time points. The matched error rate, shown in Fig. 3d and in the inset of Fig. 3c, was computed by averaging the instantaneous decoding error over all mismatched time points (that is, $\langle (s_t - \hat{s}_t)^2 \rangle_{t: \hat{c}_t \neq c_t}$). The product of these two terms was used to compute the mismatched error in Fig. 3b,e. The matched error in Fig. 3b,e was computed in an analogous manner, and the sum of matched and mismatched errors was used to compute the global errors in Figs. 1d and 3b,f. For visualization purposes, the error values in Fig. 3d–f (corresponding to $\alpha_{\text{NL}} > 0$ and $\alpha_i > 0$) were smoothed using a boxcar filter of width 8. This smoothing was done purely for display purposes; the optimal values of α_i and α_{NL} were selected before smoothing.

Mean and variance adaptation in single neurons. *Adaptive encoder.* We modeled a single neuron whose response r_t to a stimulus s_t was determined by non-linear function $f(s_t) = 1/(1 + \exp(-k(s_t - s_0)))$, where the adaptable encoding parameters $\psi = \{k, s_0\}$ specify the slope and offset of the non-linearity, respectively. We added Gaussian noise with standard deviation $\sigma^2 = 0.01$ to the output of this non-linearity. We then thresholded values below 0 or above 1, and we discretized the output into n response levels of equal width, ranging from 0 to $n - 1$. We took the discrete value at time t (representing a spike count) to be the neural response r_t . All results were generated with $n = 8$ response levels.

Model environment. We considered an environment consisting of two contexts: a 'low' context ($c = c_L$) and a 'high' context ($c = c_H$). The environment could switch between these two contexts at a small but fixed probability per time, p_s (that is, $p(c_t|c_{t-1}) = p_s$ if $c_t \neq c_{t-1}$ and $(1 - p_s)$ otherwise). Unless otherwise specified, all results were generated with $p_s = 0.01$ (see below for details of simulations that vary this probability). We considered two environments, so-called 'mean adaptation' and 'variance adaptation', in which the context c parameterized either the mean or the standard deviation of a Gaussian stimulus distribution. In the former case, we took $p(s|c) = \mathcal{N}(s; c, 1)$, with $c_L = -2$ and $c_H = 2$. In the latter case, we took $p(s|c) = \mathcal{N}(s; 0, c^2)$, with $c_L = 1$ and $c_H = 3$.

Sensory task. We evaluated the performance of the system by measuring the average error in decoding stimuli, $\langle (s - \hat{s})^2 \rangle$. To compute this error, we decoded an estimate of the stimulus $\hat{s}_t = p_1 r_t + p_0$ using a linear function with an adaptable slope p_1 and offset p_0 (described in more detail below).

Observer. To aid in task performance, the system constructs an internal estimate of context $\hat{c}_t$ that is used to select the appropriate encoding parameters, and also the appropriate decoding parameters, at each time step. We used an optimal Bayesian observer to construct the estimate $\hat{c}_t$. To this end, the neural response s_i was first decoded as described above and was then used to update the posterior distribution over c_t . Because the context can take only two values, the posterior can be summarized by the probability that the environment is in the high context at time t , given by $P_t^H = p(c_t = c_H | \hat{s}_{t \leq t})$. On each time step, P_t^H was updated with incoming stimulus estimates (as originally derived in ref. ²¹ for raw stimulus values; also see ref. ⁵⁰). We averaged the posterior to compute the point estimate

$\hat{c}_t = P_t^H c_H + (1 - P_t^H) c_L$, which could then be used to estimate the stimulus distribution $p(s|\hat{c}_t)$ (where $p(s|\hat{c}) = \mathcal{N}(s; \hat{c}, 1)$ in a mean-switching environment and $p(s|\hat{c}) = \mathcal{N}(s; 0, \hat{c}^2)$ in a variance-switching environment). This estimate was fed back upstream and used to determine the optimal encoding and decoding parameters on the next time step.

Optimization of non-linear response functions. The set of encoding parameters Ψ was optimized offline for different values of the system's current posterior P_t^H . In practice, we discretized the posterior into 99 evenly spaced values between 0.01 and 0.99, and we determined the corresponding set of encoding parameters $\Psi = \{\psi_n\}$, where $n \in [1, 99]$ indexes the possible values of the posterior (and corresponding values of $\hat{c}$). During online simulations (described below), a single set of encoding (and decoding) parameters was selected from this set given the system's current estimate of context, $\hat{c}$.

For each value of the posterior (and each corresponding value of $\hat{c}$), we performed an exhaustive grid search over a 201×201 grid of parameter values, spanning the ranges $[0, 2]$, $[0, 1.5]$ for the slope k (in the mean- and variance-switching environments, respectively) and $[-2.5, 2.5]$, $[-1, 0]$ for the offset s_0 (mean- and variance-switching, respectively). We then generated 1×10^5 stimulus samples drawn from $p(s|\hat{c})$. We passed this set of stimulus samples through the encoding non-linearity, defined by a given combination of slope k and offset s_0 , to generate a distribution of spike counts r . We computed the parameters p_0 and p_1 of the optimal linear decoder that minimized the decoding error $\langle (s - \hat{s})^2 \rangle_{p(s|\hat{c})}$ between the true s and decoded $\hat{s}$ stimulus values. The resulting set of decoded stimuli was used to update the estimate $\hat{c}(\hat{s})$ and compute the average inference error $\langle (\hat{c}(s) - \hat{c}(\hat{s}))^2 \rangle_{p(s|\hat{c})}$. When iterated over all combinations of s_0 and k , this procedure resulted in two error landscapes, $E_{\text{inf}}(k, s_0 | P_t^H)$ and $E_{\text{dec}}(k, s_0 | P_t^H)$, for each value of the posterior P_t^H .

In the mean-switching environment, the symmetry of the problem guarantees that the error landscapes are equivalent under the exchange $s_0 \rightarrow (-s_0)$ and $P_t^H \rightarrow (1 - P_t^H)$. We used this equivalence to reduce numerical noise in our optimization through the following construction: $E(k, s_0 | P_t^H) = \frac{1}{2} [E(k, s_0 | P_t^H) + E(k, -s_0 | 1 - P_t^H)]$. This was not performed for the variance-switching environment. We then normalized the values of each landscape to lie between 0 and 1, and we smoothed each landscape by replacing the error at a given entry (s_0, k) with the average over a 12×12 region around the entry. We then constructed weighted combinations of the two landscapes: $E_{\alpha_i}(k, s_0 | P_t^H) = (1 - \alpha_i) E_{\text{dec}}(k, s_0 | P_t^H) + \alpha_i E_{\text{inf}}(k, s_0 | P_t^H)$, where α_i is a weighting factor that we varied between 0 (pure decoding) and 1 (pure inference). Finally, we found the combination of parameters (s_0, k) (with corresponding decoding parameters (p_0, p_1)) that minimized the error $E_{\alpha_i}(k, s_0 | P_t^H)$ for each value of P_t^H and each value of α_i .

When $\alpha_i = 0$, this procedure defines the parameters $\{\psi_{\epsilon}^{\text{local}}\}$ of the naively adaptive code. When $\alpha_i > 0$, this procedure defines the parameters $\{\psi_{\epsilon}^{\text{infer}, \alpha_i}\}$ of a set of codes that are biased toward inference and, thus, toward reducing the duration of mismatch. We empirically found that this form of biasing led to the lowest global error. We used 101 values of α_i spaced evenly between 0 and 1. In the main text, results are shown for $\alpha_i = [0, .02, .04, .06, .08, .1, .2, .3, .4, .5, .6, .7, .8, .9, 1]$.

An analogous procedure was used to determine the parameters ψ^{global} and $\{\psi_{\epsilon}^{\text{local}}, \psi_{\epsilon}^{\text{local}}\}$ of the static and oracle codes, respectively. In the former case, we uniformly sampled 1×10^5 stimulus values from both contexts (that is, we sampled 5×10^4 samples from each context). We used the grid search described above to determine the encoding and decoding parameters that minimize the decoding error $\langle (s - \hat{s})^2 \rangle_{p(s)}$ averaged over the distribution of stimuli marginalized over both contexts (that is, $p(s) = \frac{1}{2} (p(s|c_L) + p(s|c_H))$). In the latter case, we sampled 1×10^5 stimulus values from each context and used the same grid search to determine the encoding and decoding parameters that separately minimize the decoding errors $\langle (s - \hat{s})^2 \rangle_{p(s|c_L)}$ and $\langle (s - \hat{s})^2 \rangle_{p(s|c_H)}$.

Simulations. We simulated a probe environment that switched between c_L and c_H every $1/p_s = 100$ time steps; this probe signal is not unlikely given the generative process for c_t ²¹. At each time step t , the system maintained a posterior belief P_{t-1}^H and corresponding point estimate $\hat{c}_{t-1}$ from the previous time step. The posterior belief specified the set of optimal encoding and decoding parameters to be used on the current time step (as described above). We then randomly sampled a single stimulus sample s_i from $p(s|c_t)$. This stimulus sample was encoded and decoded to construct an estimate $\hat{s}_t$, which was then used to update the posterior belief and compute the instantaneous decoding error $(s_t - \hat{s}_t)^2$.

We defined a single trial of the probe environment to consist of 100 time steps in the low state ($c_t = c_L$), followed by 100 time steps in the high state ($c_t = c_H$). We simulated dynamics over 5,000 continuous trials, and we averaged the results across trials. This resulted in trial-averaged values of the posterior, context estimate, and decoding error at each point in time. These trial-averaged values (shown in Fig. 4f) were used to compute all of the remaining results in Fig. 4.

To define periods of match, we selected all time points for which $P_t^H > .95$ (when $c_t = c_H$) and for which $(1 - P_t^H) > .95$ (when $c_t = c_L$). All other time points were labeled as mismatched. The relative duration of mismatch, shown in Fig. 4g, was computed by scaling the total number of mismatched time points by the total number of all time points (that is, 200 time points). The mismatched error rate,

also shown in Fig. 4g, was computed by averaging the decoding error over all mismatched time points. The product of these two terms was then used to compute the mismatched error in Fig. 4h. The matched error in Fig. 4h was computed in an analogous manner, and the sum of matched and mismatched errors was used to compute the global error in Fig. 4i.

We separately varied the volatility of context switches by varying the switching probability p_s . All results in Figs. 4b–e and 5a,b were generated using the probe signal described above, with $p_s = 0.01$. The results in Extended Data Fig. 4 and Fig. 5c,d were generated using a probe environment with $p_s = [0.005, 0.01, 0.015, 0.02, 0.025]$ (corresponding to a period of $1/p_s = [200, 100, 66, 50, 40]$). For each value of p_s , we simulated 5,000 continuous trials of the corresponding probe environment and averaged results across trials as described above. The effect of the switching probability on the duration, error rate, and error of matches and mismatches is shown in the upper panels of Extended Data Fig. 4.

To compute the net change in duration, shown in the lower panels of Extended Data Fig. 4a, we subtracted the relative duration of the naively adaptive code from all codes. We did this separately for each value of p_s . To compute the percent change in the mismatched error rate, mismatched error, matched error, and global error (shown in the lower panels of Extended Data Fig. 4), we first subtracted the corresponding error of the naively adaptive code from all codes, and we then scaled the resulting change in error by the difference in error between the static and oracle code.

The results shown in Fig. 4b–e summarize the behavior of the optimal non-linearities $\{\psi_{\hat{c}}^{\text{local}}\}$ (for $\alpha_i = 0$) and $\{\psi_{\hat{c}}^{\text{infer}, \alpha_i}\}$ (for $\alpha_i > 0$) for different values of P_i^H (and, thus, for different values of $\hat{c}$). For each possible value of P_i^H (as described above, we discretized the posterior into 99 values in total), we used the corresponding values of k, s_i to compute the encoding non-linearity. We discretized the output of this non-linearity into $n = 8$ discrete response levels (as described above), and we then determined the $n - 1$ values along the stimulus axis that, when passed through this non-linearity, marked the transitions between these n response levels. Figure 4d,e shows these $n - 1$ stimulus values as a function of P_i^H .

Entropy coding. In Fig. 5a,b, we used an entropy coding procedure to recode the discrete output of the encoding non-linearity (see also Extended Data Fig. 5). Entropy coding assigns codewords to response levels based on the predicted probability that the response level will be used, such that shortest codewords (lowest spike counts) are assigned to the most probable levels. In our case, we used entropy coding to reassign spike counts $[0, \dots, n - 1]$ to the n response levels of the encoding non-linearity based on the system's current estimate of the stimulus distribution, $p(s_i | \hat{c}_{i-1})$. We performed this recoding for each time point and on each individual trial. We first sampled 5×10^4 stimulus samples from this estimated stimulus distribution. We then passed these stimulus samples through the encoding non-linearity specified by this same context estimate, and we binned the output into the n response levels (Extended Data Fig. 5a). This resulted in a histogram of counts that specified how often each response level was expected to be used (Extended Data Fig. 5b, left). The response level that had the highest count was assigned zero spikes. The remaining response levels were sorted in descending order according to their count and were assigned the remaining $[1, \dots, n - 1]$ spikes (Extended Data Fig. 5b, center). This resulted in a recoded histogram for which the fewest spikes are expected to be used with the greatest frequency (Extended Data Fig. 5b, right). We then used this mapping to assign a spike count r_i to the stimulus sample s_i that was drawn at the given time point and on the given trial (Extended Data Fig. 5c). This resulted in a new recoded raster of spike counts that we used to compute the trial-averaged firing rates shown in the left columns of Fig. 5a,b (Extended Data Fig. 5d,e shows the firing rates before and after recoding).

We compared the firing rates in the left column of Fig. 5a to data from adapting and sensitizing ganglion cells in the retina of salamander and mice, responding to step increases in contrast (figure reproduced from Fig. 1a in ref. ²). We compared the firing rates in the left column of Fig. 5b to data from two ON ganglion cells in mouse retina, both responding to the same two step increases in luminance (figure reproduced from Fig. 4 in ref. ³⁵, using ND6 filters).

In Fig. 5c,d, we computed the timescale of these firing rate dynamics after an increase in stimulus variance (Fig. 5c) or mean (Fig. 5d). We first performed the same entropy recoding described above to the simulations generated for different values of p_s . We then fit an exponential of the form $a \exp(-t/b) + c$ to the resulting trial-averaged firing rates. To perform this fitting, we used the first 40 time points (equal to the period of the fastest context switches) after an increase in variance or mean. For large values of α_i , there was a slower initial increase in firing rate after an increase in variance, followed by exponential decay. We, thus, first determined the time Δt until the firing rate crossed a threshold of $BL + 0.5(FR_{\text{max}} - BL)$, where FR_{max} is the maximum firing rate, and BL is the baseline firing rate determined from the final five time points before a decrease in mean or variance. We removed time points $t < \Delta t$, performed the exponential fitting on the remaining time points, and computed the total timescale $\tau = b + (\Delta t - \langle \Delta t \rangle)$ (where $\langle \Delta t \rangle$ is averaged over different values of p_s for a given value of α_i). The term $(\Delta t - \langle \Delta t \rangle)$ was only non-zero for large values of α_i in response to changes in variance; in all other cases, $\Delta t = 1$ for all values of p_s , and, thus, this term was 0. This procedure is qualitatively similar to the fitting used in ref. ³² and shown in Fig. 5c,d (reproduced from Figs. 1e and 2e in ref. ³²).

To summarize the behavior across different values of α_i , we fit a line to the relationship between the timescale τ and the switching period $1/p_s$. Examples of these best-fitting lines are shown in the insets of Fig. 5c,d for different values of α_i .

Adaptation of receptive fields. Adaptive encoder. We modeled a population of two neurons that jointly encode image patches sampled from natural scenes. This model falls within a broader class of linear neural models that have been used to both capture coding properties of real neurons⁵³ and predict coding properties from theoretical principles³⁷. An incoming image patch s_i was transformed into a response vector $\mathbf{r}_i = [r_1, r_2]_i$ via a pair of adaptable receptive fields (implemented as adaptable linear filters). This response vector was then used both to perform the sensory task and to update an internal estimate of context, as described below.

Model environment. We considered an environment consisting of four contexts c . The environment could switch between these four contexts at a small but fixed probability per time, p_s (that is, $p(c_i | c_{i-1}) = p_s/3$ if $c_i \neq c_{i-1}$ and $(1 - p_s)$ otherwise). All results were generated with $p_s = 0.01$.

To define the set of contexts, we first sampled 2×10^5 image patches of 12×12 pixels from standardized images taken from a natural movie dataset, as described above. We then performed principal component analysis on this set of images. We retained the first two principal components, to which we applied a k -means clustering algorithm (using the built-in MATLAB function `kmeans.m`) to define four distinct clusters. We then used these clusters to assign individual image patches to one of these four contexts.

Sensory task. We evaluated the performance of the system on the task of maximizing information transmission. As specified by the logistic ICA model³⁷, information maximization is achieved by obtaining maximally independent filter outputs with logistic marginal distributions (see below).

To measure task error, we computed the distance between the observed distribution of filter outputs and the information-maximizing distribution of filter outputs over a time window ΔT . To simplify this computation, we summarized the observed distribution of filter outputs over the period $[t - \Delta T + 1, t]$ by a normal distribution $p_i(r_1, r_2) = \mathcal{N}(\mu_i, \Sigma_i)$, where μ_i and Σ_i are the 2×1 vector of means and 2×2 matrix of covariances, computed over the interval ΔT . Because the mean filter output does not affect information transmission, we subtracted this before evaluating the error. The Gaussian approximation of the information-maximizing distribution was defined as $p^G(r_1, r_2) = \mathcal{N}(\mu^G, \Sigma^G)$, where $\mu^G = [0, 0]$ is the vector of means and Σ^G is the diagonal covariance matrix. The diagonal entries in Σ^G were equal to 3.29—that is, the variance of the logistic distribution with a unitary scale parameter, as assumed by the ICA model³⁷ (see below). We quantified the distance between $p_i(r_1, r_2)$ and $p^G(r_1, r_2)$ using the symmetrized KL divergence, $D_{\text{KL}}(p_i \| p^G) + D_{\text{KL}}(p^G \| p_i)$, and we used this KL error as our measure of task error. We use the symmetrized, rather than non-symmetrized, KL divergence because it conceptually matches the other (symmetric) error measures discussed here.

Observer. To aid in task performance, the system constructs an internal estimate of the current context from the output of the receptive fields. To compute this estimate, we used a Gaussian mixture model (GMM) to approximate the distribution of filter outputs within each context. The parameters of the GMM were defined by the mean filter output within each context and the covariance in filter outputs across all four contexts. The context estimate $\hat{c}$ was taken to be the context that maximizes the likelihood of filter outputs given the parameters of the GMM.

Optimization of receptive fields. To obtain model receptive fields, we performed ICA³⁷ on image patches belonging to each individual context. Because our goal was to learn a set of two receptive fields, we reduced the dimensionality of image patches to the first two principal components (as described above) before applying ICA. This resulted in a pair of context-optimal receptive fields for each of the four contexts. A single set of global receptive fields was obtained in the same manner, by performing ICA on the first two principal components of the ensemble of image patches across all four contexts.

We used ICA to learn context-specific filters that were biased toward minimizing the mismatched error rate. For each context, we created a training dataset in which $(1 - \alpha_{\text{NL}})N$ patches were sampled from the current context and $\alpha_{\text{NL}}N$ patches were sampled uniformly from all other contexts, where $N = 10^4$ is the total number of patches. We then trained ICA filters on that new dataset. We considered ten values of α_{NL} spaced linearly on the interval $[0, 0.625]$. We found empirically that this biasing strategy resulted in the largest decrease of the total KL error with respect to the naively adaptive code ($\alpha_{\text{NL}} = 0$).

Simulations. We simulated dynamics over 2×10^5 time steps. At each time step t , the system maintained a context estimate $\hat{c}_{t-1}$ from the previous time step. This estimate was used to select the set of receptive fields to be used on the current time step. We then randomly sampled a single image patch from the ensemble of patches specified by the current context c_t . This image patch was encoded in a response $\mathbf{r}_t$ as described above, which was then used to update the context estimate $\hat{c}_t$. The KL error was computed from these responses every ten time steps.

over intervals of $\Delta T = 1/p_i = 100$ time steps. A given interval was defined to be 'matched' if at least 80% of the context estimates from that interval were correctly classified via the GMM inference process described above. Otherwise, the interval was defined to be 'mismatched'. To compute the matched and mismatched errors shown in Fig. 6e, we first summed the KL error per interval over all matched and mismatched intervals, respectively, and then scaled these by the total number of all intervals. We then summed these two errors to compute the global errors shown in Fig. 6g. Classification error, shown in Fig. 6f, was estimated by fitting the GMM to outputs of context-specific filters specified by each value of α_{NL} and computing the misclassification rate for individual stimulus samples (image patches).

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Data sharing is not applicable as no datasets were generated or analyzed in the current study.

Code availability

Code was written in MATLAB and is available at <https://github.com/HermundstadLab/EfficientAdaptiveCodes>.

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Acknowledgements

We thank D. Kastner and T. Münch for generously providing figures from their work. We also thank V. Jayaraman, M. Noorman, T. Ma, and K. Krishnamurthy for useful discussions and feedback on the manuscript. W.F.M. was funded by the European Union's Horizon 2020 Research and Innovation Programme under Marie Skłodowska-Curie Grant Agreement No. 754411. A.M.H. was supported by the Howard Hughes Medical Institute.

Author contributions

W.F.M. and A.M.H. conceived the study, developed the theoretical framework, conducted the mathematical analysis, performed the simulations, interpreted the results, and wrote the paper.

Competing interests

The authors declare no competing financial interests.

Additional information

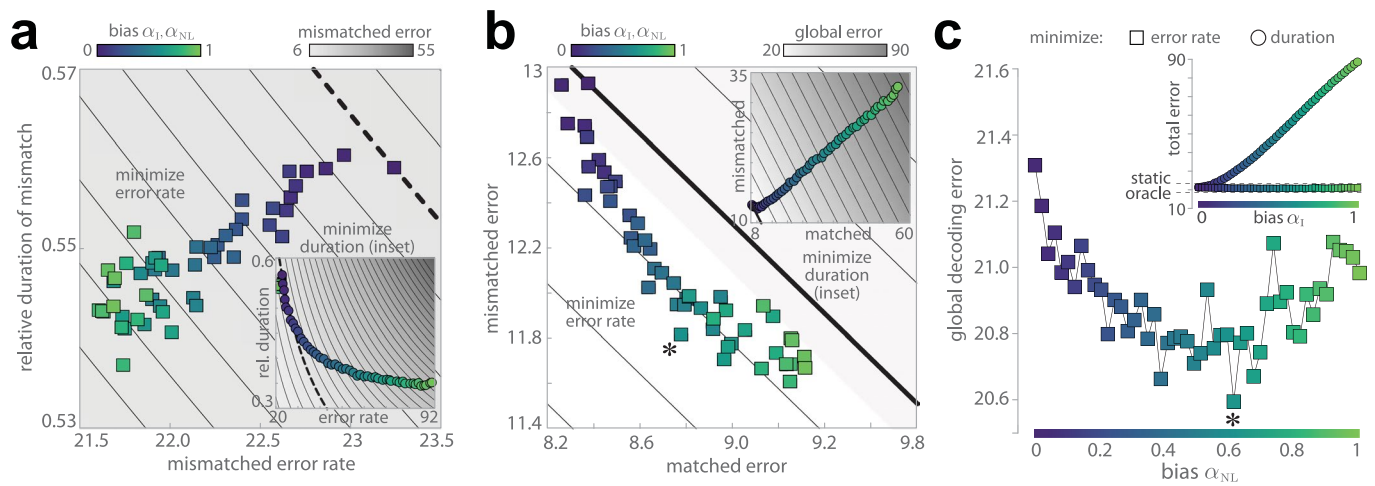
Extended data is available for this paper at <https://doi.org/10.1038/s41593-021-00846-0>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41593-021-00846-0>.

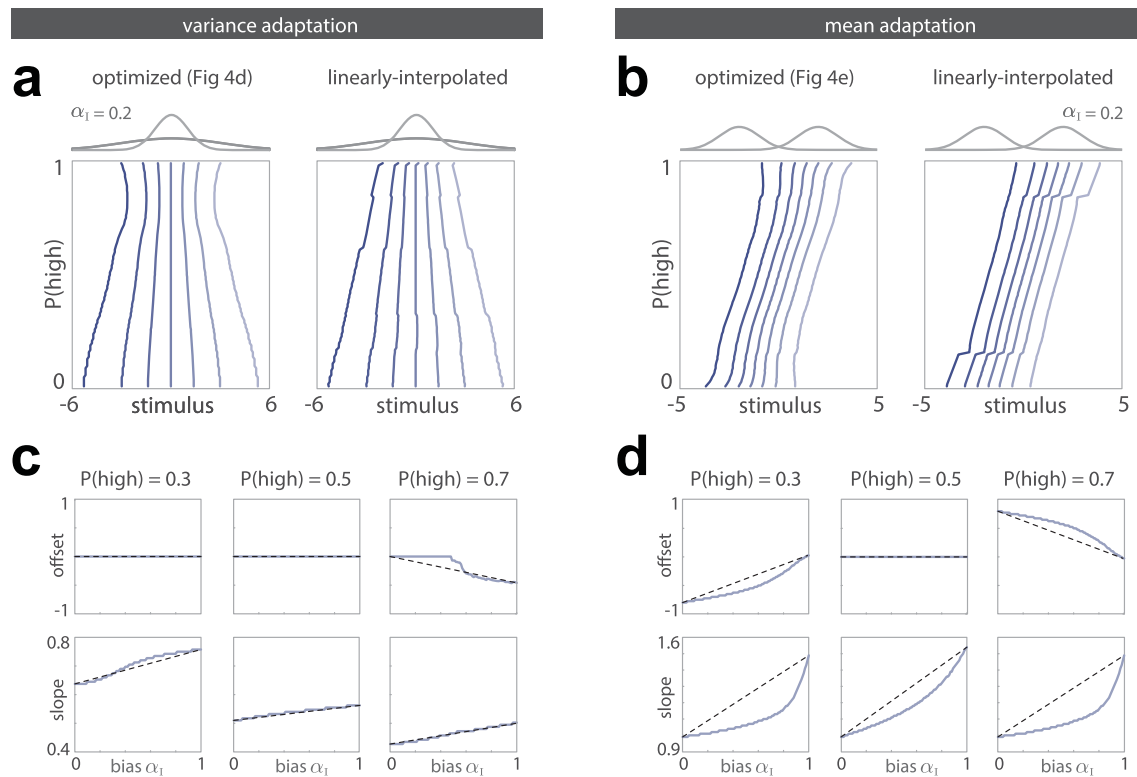
Correspondence and requests for materials should be addressed to W.F.M. or A.M.H.

Peer review information *Nature Neuroscience* thanks the anonymous reviewers for their contribution to the peer review of this work.

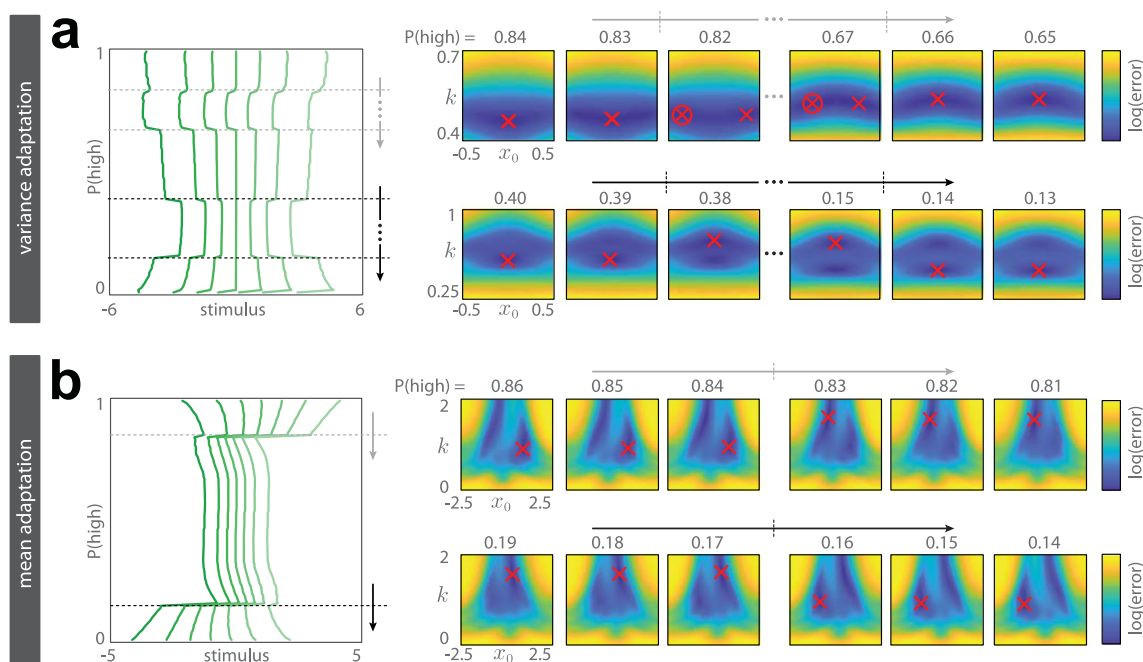
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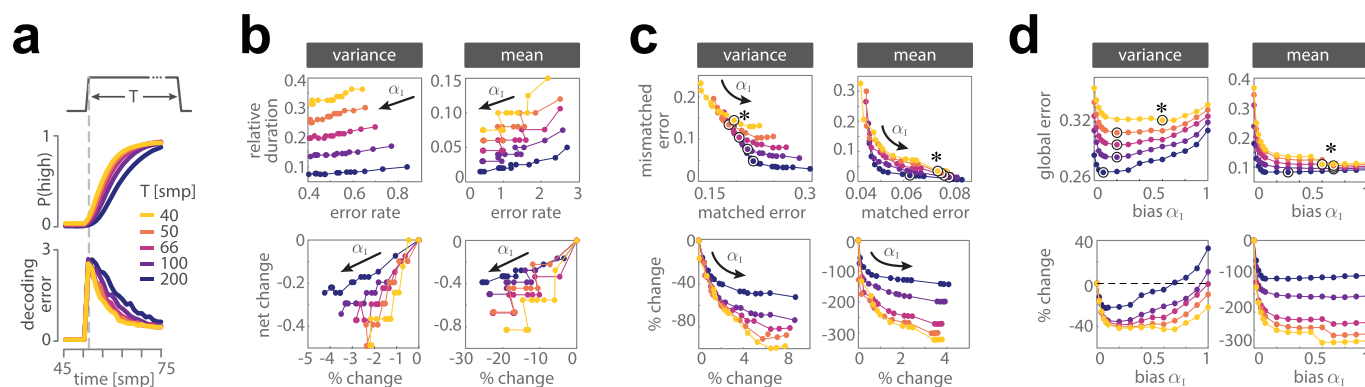
Extended Data Fig. 1 | Smoothing does not alter results. a–c, Same results as shown in Fig. 3d–f, here without smoothing. Optimal values of α_{NL} (stars) were chosen prior to smoothing.



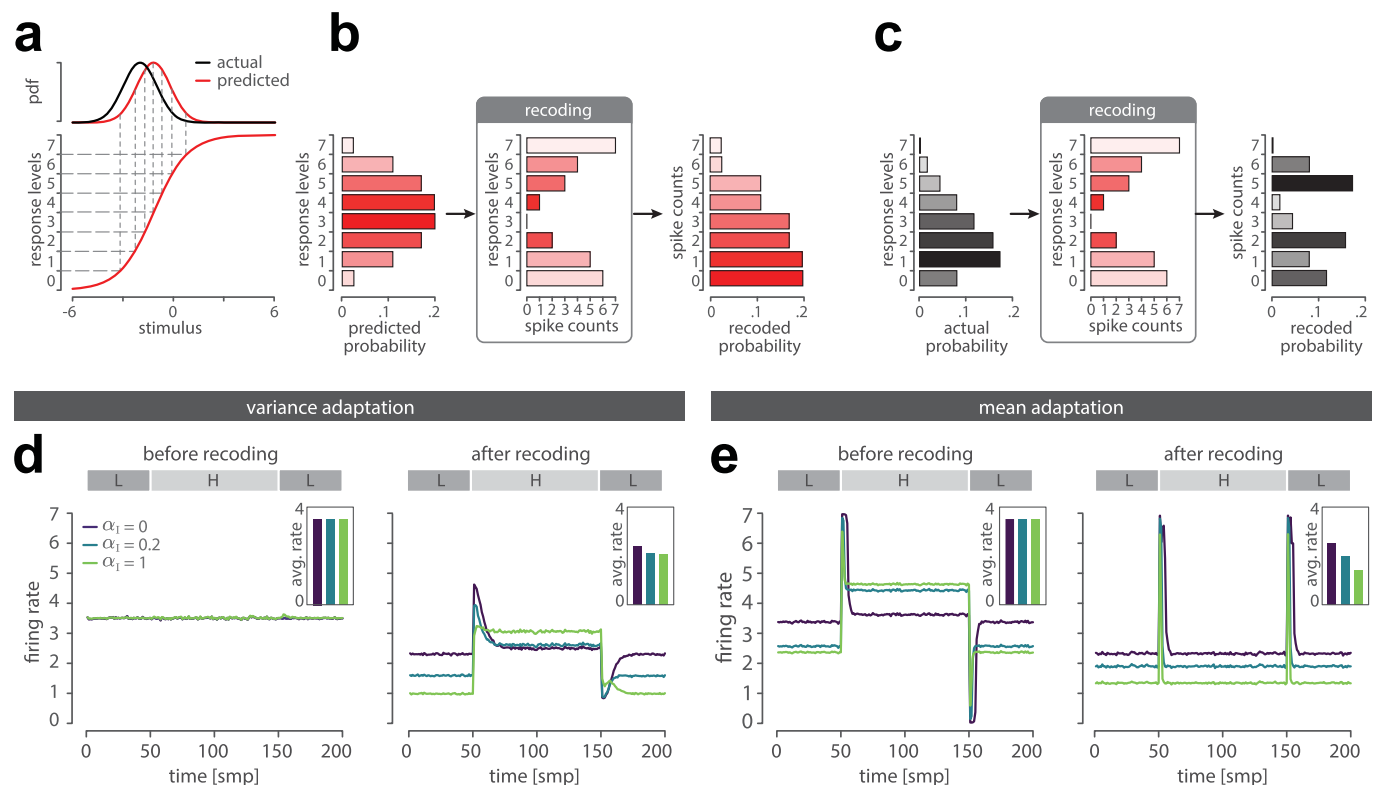
Extended Data Fig. 2 | Optimal dynamics are robust to parameter variations. **a, b**, Optimal nonlinearities (left; shown in Fig. 4d,e) closely resemble those derived by linearly-interpolating parameters between the naively-adaptive code and the fully-biased code (right). Shown here as a function of the system's posterior belief $P(\text{high}) = P_t^H$, for variance adaptation (**a**) and mean adaptation (**b**). **c, d**, The dependence of the offset (upper row) and slope (lower row) of the optimal nonlinearities on the bias α_1 can be approximated by a line that interpolates between the naively-adaptive code ($\alpha_1 = 0$) and the fully-based code ($\alpha_1 = 1$). Shown here for variance adaptation (**c**) and mean adaptation (**d**), for several different values of the posterior belief $P(\text{high})$.



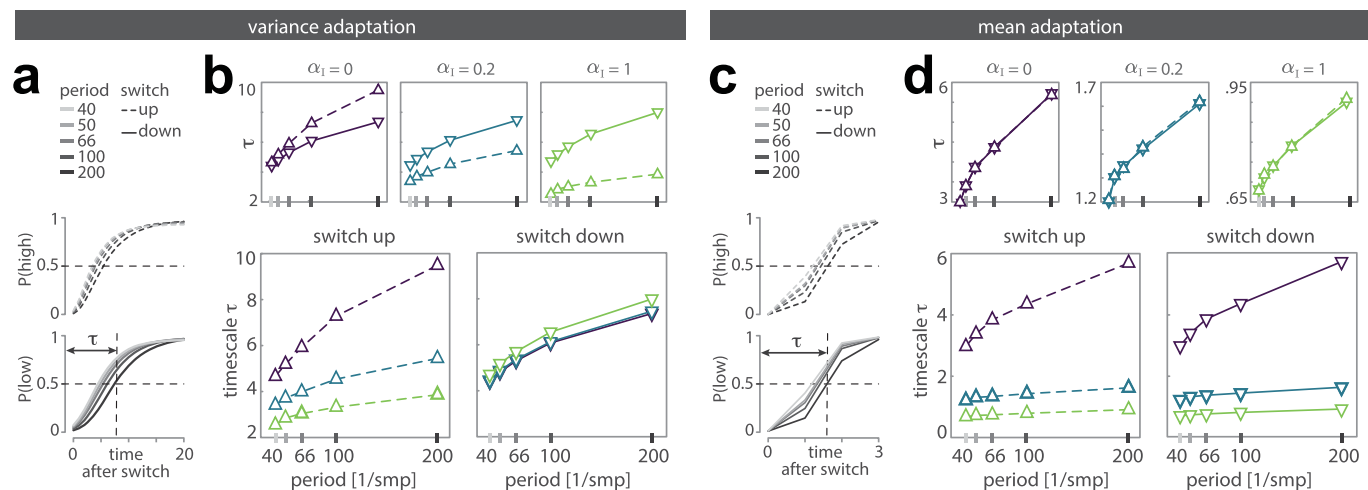
Extended Data Fig. 3 | Abrupt changes in encoding parameters arise from bifurcations in error landscapes. **a,b**, Encoding nonlinearities (left panels; shown in Fig. 4d,e) show abrupt changes for small variations in the system's posterior belief $P(\text{high}) = P_t^H$. These changes arise from variations in the underlying error landscapes (right panels) in which a single minimum bifurcates into two minima or vice-versa (**a**, upper row; global minima marked by red x's), or when one of multiple local minima becomes the global minimum (**a**, lower row; **b**, both rows). In the case of variance adaptation, the error landscape is symmetric to positive versus negative variations in the offset of the encoding nonlinearity (x_0), and thus bifurcations in the error landscape lead to two symmetric solutions (**a**, upper row). In such scenarios, we chose the solution corresponding to the lower value of x_0 (marked by circles), because it produces lower spike counts on average.



Extended Data Fig. 4 | Balance of errors is robust to changes in volatility. **a**, As volatility increases, the posterior is faster to update and the error faster to decay (shown for the naively-adaptive code in response to an increase in variance). **b–d**, Biased codes produce qualitatively consistent patterns of performance regardless of volatility (upper panels), but the relative impact of biasing the code increases as volatility increases (lower panels). Higher volatility leads to relatively longer periods of mismatch (**b**, upper), higher mismatched errors and lower matched errors (**c**, upper), and higher global errors (**d**, upper). The percent change in each of these errors (measured relative to the naively-adaptive code and scaled by the difference in error between the static and oracle codes) increases as volatility increases (lower row), and the global error is thus minimized for larger values of bias (**c–d**; circled markers in upper panels).



Extended Data Fig. 5 | Entropy coding transforms response levels and reduces average firing rates. a–c. Entropy coding reassigns response levels to spike counts based on the predicted probability that a response level will be used. **a**, The encoding nonlinearity partitions stimuli drawn from the actual stimulus distribution (black) or predicted stimulus distribution (red) into discrete response levels. This partitioning determines the predicted (**b**, left) and actual (**c**, left) probability with which response levels will be used. **b**, Entropy coding reassigns spike counts in order of decreasing predicted probability (left); response levels that have higher predicted probability (darker red) will be assigned fewer spikes (middle). This leads to a recoded histogram that is weighted toward lower spike counts (right). The reassignment could be approximated by a quadratic nonlinearity, or by a thresholding exponential nonlinearity. **c**, The recoding scheme, which is based on *predicted* probability (**b**), determines how stimuli sampled from the *actual* stimulus distribution will be transformed into spike counts. **d**, Before recoding (left), firing rates do not change in response to a change between low (L) and high (H) stimulus variance, regardless of the value of α_1 . After recoding (right), all codes show a transient response to both increases and decreases in variance, and time-averaged firing rates are lower (inset). **e**, Before recoding (left), firing rates increase following a change from low (L) to high (H) stimulus mean, and decrease following a change from high to low mean. After recoding (right), all codes show a symmetric response to increases and decreases in mean, and time-averaged firing rates are lower (inset).



Extended Data Fig. 6 | Volatility shapes posterior dynamics. a, c, As volatility increases (that is, as the switching period decreases), the posterior is faster to respond to both increases (upper) and decreases (lower) in stimulus variance (**a**) and mean (**c**). We estimated this timescale by linearly interpolating the posterior before computing the time at which it crosses 0.5. **b,** When stimulus variance is changing in time, the naively-adaptive code ($\alpha_1 = 0$, upper left) is faster to respond to decreases in variance (solid lines) than to increases (dashed lines). In contrast, biased codes (upper middle, right) are faster to respond to increases in variance. This is largely driven by a more rapid response to increases in variance with increasing bias (lower left), but is also weakly affected by a slower response to decreases in variance (lower right). **d,** The posterior is equally fast to respond to increases and decreases in the stimulus mean (upper row), with a timescale that decreases as bias increases (lower row).

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No data was collected during this study

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All code was written in Matlab R2020a. Code is available at <https://github.com/HermundstadLab/EfficientAdaptiveCodes>.

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